# Online Resource 1

## Characterizing Scanpath Decoding Pipelines with Perturbation Profiles

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Tables S1–S13: every profile of the primary analysis (within-fold AUC, class-balanced participant folds); $R = 6$ replicates, $B = 60$ refit-bootstrap draws, seed 42; linear SVM, ten-fold participant-held-out cross-validation; duration-marginal donor pool of the other participants' durations. $u$ is the reference scale (one text line in SB-SAT and EMIP, 2° in GazeBase). The same profiles with the pooled AUC are in the released outputs.

Table S1: Profile of `gb_tex_vd1`: reading vs. video viewing. Sensitivity (AUC drop) with refit-bootstrap 95% interval; bold: interval excludes zero.

| Descriptor | AUC | order | dur. coupl. | dur. marg. | rotation | jit. $0.25u$ | jit. $0.5u$ | jit. $1u$ | jit. $2u$ |
|---|---|---|---|---|---|---|---|---|---|
| Count | 0.969 | 0 (ex.) | 0 (ex.) | 0 (ex.) | 0 (ex.) | 0 (ex.) | 0 (ex.) | 0 (ex.) | 0 (ex.) |
| Dispersion | 1.000 | 0 (ex.) | 0 (ex.) | 0 (ex.) | **+0.021 [+0.01, +0.03]** | +0.000 [−0.00, +0.00] | −0.000 [−0.00, +0.00] | +0.000 [−0.00, +0.00] | +0.000 [−0.00, +0.00] |
| Classic | 0.969 | **−0.022 [−0.03, −0.01]** | 0 (ex.) | **+0.035 [+0.02, +0.05]** | 0 (ex.) | **−0.003 [−0.01, −0.00]** | **−0.006 [−0.01, −0.00]** | **−0.007 [−0.01, −0.00]** | −0.004 [−0.01, +0.00] |
| Dur. trend | 0.957 | +0.003 [−0.00, +0.01] | +0.003 [−0.00, +0.01] | **+0.473 [+0.44, +0.50]** | 0 (ex.) | 0 (ex.) | 0 (ex.) | 0 (ex.) | 0 (ex.) |
| RQA $0.5u$ | 0.999 | **+0.010 [+0.00, +0.02]** | 0 (ex.) | 0 (ex.) | 0 (ex.) | +0.001 [−0.00, +0.00] | +0.002 [−0.00, +0.00] | **+0.004 [+0.00, +0.01]** | **+0.015 [+0.01, +0.02]** |
| RQA $1u$ | 1.000 | **+0.006 [+0.00, +0.01]** | 0 (ex.) | 0 (ex.) | 0 (ex.) | +0.000 [−0.00, +0.00] | +0.001 [−0.00, +0.00] | +0.001 [−0.00, +0.00] | **+0.006 [+0.00, +0.01]** |
| Directed | 0.977 | **−0.014 [−0.02, −0.01]** | 0 (ex.) | 0 (ex.) | **−0.007 [−0.01, −0.00]** | **−0.011 [−0.02, −0.00]** | **−0.015 [−0.02, −0.01]** | **−0.016 [−0.02, −0.01]** | **−0.013 [−0.02, −0.01]** |
| FAG lattice | 0.988 | 0 (ex.) | −0.001 [−0.00, +0.00] | +0.004 [−0.00, +0.01] | +0.002 [−0.00, +0.01] | −0.000 [−0.01, +0.00] | +0.001 [−0.00, +0.01] | **+0.009 [+0.00, +0.02]** | **+0.049 [+0.04, +0.06]** |

Table S2: Profile of `gb_vd1_blg`: video viewing vs. ball-catching game. Sensitivity (AUC drop) with refit-bootstrap 95% interval; bold: interval excludes zero.

| Descriptor | AUC | order | dur. coupl. | dur. marg. | rotation | jit. $0.25u$ | jit. $0.5u$ | jit. $1u$ | jit. $2u$ |
|---|---|---|---|---|---|---|---|---|---|
| Count | 0.690 | 0 (ex.) | 0 (ex.) | 0 (ex.) | 0 (ex.) | 0 (ex.) | 0 (ex.) | 0 (ex.) | 0 (ex.) |
| Dispersion | 1.000 | 0 (ex.) | 0 (ex.) | 0 (ex.) | +0.000 [−0.00, +0.00] | +0.000 [−0.00, +0.00] | +0.000 [−0.00, +0.00] | −0.000 [−0.00, +0.00] | +0.000 [−0.00, +0.00] |
| Classic | 0.886 | **−0.114 [−0.14, −0.09]** | 0 (ex.) | **+0.033 [+0.02, +0.05]** | 0 (ex.) | **−0.003 [−0.00, −0.00]** | **−0.007 [−0.01, −0.00]** | **−0.014 [−0.02, −0.01]** | **−0.014 [−0.02, −0.01]** |
| Dur. trend | 0.752 | +0.011 [−0.00, +0.02] | +0.011 [−0.00, +0.02] | **+0.252 [+0.20, +0.30]** | 0 (ex.) | 0 (ex.) | 0 (ex.) | 0 (ex.) | 0 (ex.) |
| RQA $0.5u$ | 0.999 | **+0.013 [+0.01, +0.02]** | 0 (ex.) | 0 (ex.) | 0 (ex.) | −0.000 [−0.00, +0.00] | −0.000 [−0.00, +0.00] | −0.000 [−0.00, +0.00] | **+0.003 [+0.00, +0.01]** |
| RQA $1u$ | 1.000 | **+0.001 [+0.00, +0.00]** | 0 (ex.) | 0 (ex.) | 0 (ex.) | −0.000 [−0.00, +0.00] | −0.000 [−0.00, +0.00] | +0.000 [−0.00, +0.00] | +0.000 [−0.00, +0.00] |
| Directed | 1.000 | **+0.005 [+0.00, +0.01]** | 0 (ex.) | 0 (ex.) | +0.000 [−0.00, +0.00] | +0.000 [−0.00, +0.00] | +0.000 [−0.00, +0.00] | +0.000 [−0.00, +0.00] | **+0.002 [+0.00, +0.00]** |
| FAG lattice | 0.957 | 0 (ex.) | +0.000 [−0.00, +0.00] | **−0.008 [−0.01, −0.00]** | **+0.029 [+0.02, +0.04]** | +0.007 [−0.00, +0.02] | +0.010 [−0.00, +0.02] | **+0.019 [+0.00, +0.04]** | **+0.027 [+0.01, +0.04]** |

Table S3: Profile of `gb_vd1_vd2`: video segment 1 vs. 2. Sensitivity (AUC drop) with refit-bootstrap 95% interval; bold: interval excludes zero.

| Descriptor | AUC | order | dur. coupl. | dur. marg. | rotation | jit. 0.25$u$ | jit. 0.5$u$ | jit. 1$u$ | jit. 2$u$ |
|---|---|---|---|---|---|---|---|---|---|
| Count | 0.634 | 0 (ex.) | 0 (ex.) | 0 (ex.) | 0 (ex.) | 0 (ex.) | 0 (ex.) | 0 (ex.) | 0 (ex.) |
| Dispersion | 0.707 | 0 (ex.) | 0 (ex.) | 0 (ex.) | **+0.098 [+0.07, +0.13]** | +0.004 [−0.00, +0.01] | **+0.012 [+0.00, +0.02]** | **+0.045 [+0.03, +0.06]** | **+0.116 [+0.08, +0.15]** |
| Classic | 0.640 | +0.007 [−0.00, +0.01] | 0 (ex.) | **+0.124 [+0.10, +0.15]** | 0 (ex.) | +0.001 [−0.00, +0.00] | **+0.004 [+0.00, +0.01]** | +0.005 [−0.00, +0.01] | +0.005 [−0.00, +0.01] |
| Dur. trend | 0.790 | **+0.157 [+0.13, +0.19]** | **+0.158 [+0.13, +0.19]** | **+0.286 [+0.23, +0.34]** | 0 (ex.) | 0 (ex.) | 0 (ex.) | 0 (ex.) | 0 (ex.) |
| RQA 0.5$u$ | 0.677 | **+0.035 [+0.01, +0.06]** | 0 (ex.) | 0 (ex.) | 0 (ex.) | −0.006 [−0.03, +0.01] | +0.002 [−0.03, +0.03] | **+0.046 [+0.00, +0.09]** | **+0.108 [+0.07, +0.14]** |
| RQA 1$u$ | 0.719 | **+0.073 [+0.04, +0.10]** | 0 (ex.) | 0 (ex.) | 0 (ex.) | +0.007 [−0.01, +0.02] | +0.013 [−0.01, +0.03] | **+0.043 [+0.01, +0.08]** | **+0.118 [+0.08, +0.16]** |
| Directed | 0.700 | **+0.040 [+0.02, +0.06]** | 0 (ex.) | 0 (ex.) | −0.010 [−0.03, +0.01] | −0.005 [−0.03, +0.02] | +0.013 [−0.01, +0.04] | **+0.042 [+0.01, +0.08]** | **+0.069 [+0.04, +0.10]** |
| FAG lattice | 0.664 | 0 (ex.) | **−0.015 [−0.03, −0.00]** | **−0.020 [−0.03, −0.01]** | −0.012 [−0.05, +0.02] | −0.015 [−0.05, +0.02] | −0.006 [−0.05, +0.04] | +0.017 [−0.03, +0.06] | **+0.055 [+0.01, +0.10]** |

Table S4: Profile of `gb_vd1_halves`: video 1, first vs. second half. Sensitivity (AUC drop) with refit-bootstrap 95% interval; bold: interval excludes zero.

| Descriptor | AUC | order | dur. coupl. | dur. marg. | rotation | jit. 0.25$u$ | jit. 0.5$u$ | jit. 1$u$ | jit. 2$u$ |
|---|---|---|---|---|---|---|---|---|---|
| Count | 0.500 | 0 (ex.) | 0 (ex.) | 0 (ex.) | 0 (ex.) | 0 (ex.) | 0 (ex.) | 0 (ex.) | 0 (ex.) |
| Dispersion | 0.924 | 0 (ex.) | 0 (ex.) | 0 (ex.) | **+0.249 [+0.22, +0.27]** | +0.001 [−0.00, +0.00] | **+0.007 [+0.00, +0.01]** | **+0.029 [+0.02, +0.04]** | **+0.130 [+0.11, +0.15]** |
| Classic | 0.564 | **−0.090 [−0.11, −0.07]** | 0 (ex.) | +0.008 [−0.01, +0.03] | 0 (ex.) | −0.001 [−0.00, +0.00] | +0.002 [−0.00, +0.01] | +0.005 [−0.01, +0.02] | +0.015 [−0.00, +0.03] |
| Dur. trend | 0.590 | **+0.076 [+0.03, +0.12]** | **+0.085 [+0.04, +0.13]** | **+0.082 [+0.04, +0.12]** | 0 (ex.) | 0 (ex.) | 0 (ex.) | 0 (ex.) | 0 (ex.) |
| RQA 0.5$u$ | 0.743 | **+0.046 [+0.02, +0.07]** | 0 (ex.) | 0 (ex.) | 0 (ex.) | +0.012 [−0.01, +0.03] | +0.025 [−0.00, +0.05] | **+0.079 [+0.05, +0.11]** | **+0.143 [+0.10, +0.19]** |
| RQA 1$u$ | 0.823 | **+0.088 [+0.06, +0.11]** | 0 (ex.) | 0 (ex.) | 0 (ex.) | +0.003 [−0.01, +0.02] | **+0.026 [+0.01, +0.05]** | **+0.088 [+0.05, +0.12]** | **+0.182 [+0.15, +0.21]** |
| Directed | 0.805 | **+0.036 [+0.02, +0.05]** | 0 (ex.) | 0 (ex.) | −0.007 [−0.03, +0.01] | +0.010 [−0.01, +0.03] | **+0.033 [+0.01, +0.05]** | **+0.090 [+0.06, +0.12]** | **+0.191 [+0.16, +0.22]** |
| FAG lattice | 0.741 | 0 (ex.) | −0.003 [−0.01, +0.00] | +0.000 [−0.00, +0.01] | +0.023 [−0.01, +0.05] | +0.004 [−0.02, +0.03] | +0.011 [−0.02, +0.05] | +0.037 [−0.00, +0.07] | **+0.088 [+0.05, +0.12]** |

Table S5: Profile of `gb_tex_s1s2`: reading, session 1 vs. 2. Sensitivity (AUC drop) with refit-bootstrap 95% interval; bold: interval excludes zero.

| Descriptor | AUC | order | dur. coupl. | dur. marg. | rotation | jit. 0.25$u$ | jit. 0.5$u$ | jit. 1$u$ | jit. 2$u$ |
|---|---|---|---|---|---|---|---|---|---|
| Count | 0.530 | 0 (ex.) | 0 (ex.) | 0 (ex.) | 0 (ex.) | 0 (ex.) | 0 (ex.) | 0 (ex.) | 0 (ex.) |
| Dispersion | 0.519 | 0 (ex.) | 0 (ex.) | 0 (ex.) | +0.036 [−0.05, +0.12] | +0.003 [−0.01, +0.01] | −0.000 [−0.03, +0.03] | +0.011 [−0.03, +0.06] | +0.031 [−0.01, +0.07] |
| Classic | 0.539 | +0.005 [−0.02, +0.03] | 0 (ex.) | +0.031 [−0.04, +0.10] | 0 (ex.) | +0.001 [−0.00, +0.01] | −0.000 [−0.01, +0.01] | −0.005 [−0.02, +0.02] | −0.001 [−0.03, +0.03] |
| Dur. trend | 0.549 | +0.014 [−0.04, +0.07] | +0.016 [−0.03, +0.06] | +0.036 [−0.05, +0.12] | 0 (ex.) | 0 (ex.) | 0 (ex.) | 0 (ex.) | 0 (ex.) |
| RQA 0.5$u$ | 0.576 | **+0.093 [+0.01, +0.17]** | 0 (ex.) | 0 (ex.) | 0 (ex.) | +0.037 [−0.05, +0.12] | **+0.071 [+0.02, +0.12]** | +0.072 [−0.01, +0.15] | +0.062 [−0.03, +0.16] |
| RQA 1$u$ | 0.521 | +0.047 [−0.03, +0.12] | 0 (ex.) | 0 (ex.) | 0 (ex.) | −0.007 [−0.05, +0.04] | +0.017 [−0.06, +0.09] | +0.028 [−0.06, +0.12] | +0.007 [−0.08, +0.09] |
| Directed | 0.531 | +0.028 [−0.05, +0.11] | 0 (ex.) | 0 (ex.) | +0.007 [−0.06, +0.07] | +0.003 [−0.06, +0.06] | +0.020 [−0.03, +0.07] | +0.038 [−0.02, +0.10] | +0.039 [−0.03, +0.11] |
| FAG lattice | 0.591 | 0 (ex.) | +0.010 [−0.01, +0.03] | +0.014 [−0.01, +0.04] | +0.042 [−0.03, +0.11] | +0.036 [−0.04, +0.11] | +0.056 [−0.03, +0.14] | +0.067 [−0.02, +0.16] | +0.081 [−0.01, +0.17] |

Table S6: Profile of `sb_native`: native vs. non-native English readers, reading pages. Sensitivity (AUC drop) with refit-bootstrap 95% interval; bold: interval excludes zero.

| Descriptor | AUC | order | dur. coupl. | dur. marg. | rotation | jit. 0.25$u$ | jit. 0.5$u$ | jit. 1$u$ | jit. 2$u$ |
|---|---|---|---|---|---|---|---|---|---|
| Count | 0.624 | 0 (ex.) | 0 (ex.) | 0 (ex.) | 0 (ex.) | 0 (ex.) | 0 (ex.) | 0 (ex.) | 0 (ex.) |
| Dispersion | 0.545 | 0 (ex.) | 0 (ex.) | 0 (ex.) | +0.005 [−0.06, +0.07] | +0.000 [−0.01, +0.01] | +0.000 [−0.02, +0.02] | −0.003 [−0.04, +0.03] | +0.006 [−0.06, +0.07] |
| Classic | 0.689 | +0.037 [−0.08, +0.15] | 0 (ex.) | +0.017 [−0.07, +0.10] | 0 (ex.) | +0.001 [−0.00, +0.00] | +0.004 [−0.00, +0.01] | +0.012 [−0.01, +0.03] | +0.022 [−0.04, +0.08] |
| Dur. trend | 0.654 | +0.011 [−0.04, +0.06] | +0.012 [−0.03, +0.06] | **+0.144 [+0.00, +0.29]** | 0 (ex.) | 0 (ex.) | 0 (ex.) | 0 (ex.) | 0 (ex.) |
| RQA 0.5$u$ | 0.614 | +0.074 [−0.01, +0.16] | 0 (ex.) | 0 (ex.) | 0 (ex.) | +0.012 [−0.02, +0.04] | +0.029 [−0.01, +0.07] | **+0.079 [+0.01, +0.15]** | **+0.088 [+0.01, +0.16]** |
| RQA 1$u$ | 0.669 | +0.105 [−0.00, +0.21] | 0 (ex.) | 0 (ex.) | 0 (ex.) | −0.006 [−0.03, +0.02] | +0.015 [−0.02, +0.05] | **+0.086 [+0.02, +0.16]** | **+0.140 [+0.05, +0.23]** |
| Directed | 0.614 | −0.030 [−0.09, +0.03] | 0 (ex.) | 0 (ex.) | −0.004 [−0.04, +0.03] | −0.011 [−0.04, +0.02] | −0.015 [−0.06, +0.03] | −0.014 [−0.06, +0.04] | −0.004 [−0.07, +0.06] |
| FAG lattice | 0.587 | 0 (ex.) | −0.000 [−0.02, +0.02] | +0.008 [−0.02, +0.03] | −0.002 [−0.04, +0.03] | −0.004 [−0.03, +0.02] | −0.011 [−0.06, +0.03] | −0.018 [−0.07, +0.04] | −0.022 [−0.09, +0.04] |

Table S7: Profile of `sb_compr`: high vs. low passage comprehension, reading pages. Sensitivity (AUC drop) with refit-bootstrap 95% interval; bold: interval excludes zero.

| Descriptor | AUC | order | dur. coupl. | dur. marg. | rotation | jit. 0.25$u$ | jit. 0.5$u$ | jit. 1$u$ | jit. 2$u$ |
|---|---|---|---|---|---|---|---|---|---|
| Count | 0.554 | 0 (ex.) | 0 (ex.) | 0 (ex.) | 0 (ex.) | 0 (ex.) | 0 (ex.) | 0 (ex.) | 0 (ex.) |
| Dispersion | 0.413 | 0 (ex.) | 0 (ex.) | 0 (ex.) | −0.047 [−0.12, +0.03] | −0.004 [−0.02, +0.01] | −0.007 [−0.03, +0.01] | **−0.035 [−0.06, −0.01]** | −0.041 [−0.11, +0.03] |
| Classic | 0.587 | +0.010 [−0.07, +0.09] | 0 (ex.) | +0.026 [−0.09, +0.14] | 0 (ex.) | +0.000 [−0.00, +0.00] | +0.001 [−0.01, +0.01] | +0.000 [−0.01, +0.01] | +0.003 [−0.04, +0.05] |
| Dur. trend | 0.575 | −0.006 [−0.04, +0.03] | −0.007 [−0.03, +0.02] | +0.070 [−0.05, +0.19] | 0 (ex.) | 0 (ex.) | 0 (ex.) | 0 (ex.) | 0 (ex.) |
| RQA 0.5$u$ | 0.497 | +0.011 [−0.12, +0.14] | 0 (ex.) | 0 (ex.) | 0 (ex.) | +0.009 [−0.06, +0.07] | −0.003 [−0.10, +0.09] | +0.001 [−0.08, +0.08] | −0.005 [−0.11, +0.09] |
| RQA 1$u$ | 0.529 | +0.050 [−0.04, +0.14] | 0 (ex.) | 0 (ex.) | 0 (ex.) | −0.017 [−0.06, +0.03] | −0.014 [−0.07, +0.04] | +0.006 [−0.05, +0.07] | +0.032 [−0.08, +0.15] |
| Directed | 0.533 | +0.004 [−0.07, +0.08] | 0 (ex.) | 0 (ex.) | −0.010 [−0.10, +0.07] | +0.013 [−0.05, +0.07] | −0.004 [−0.08, +0.07] | +0.003 [−0.07, +0.08] | +0.010 [−0.07, +0.09] |
| FAG lattice | 0.506 | 0 (ex.) | −0.007 [−0.04, +0.02] | −0.002 [−0.03, +0.03] | −0.048 [−0.10, +0.00] | −0.032 [−0.10, +0.04] | −0.035 [−0.11, +0.04] | −0.038 [−0.13, +0.05] | −0.011 [−0.10, +0.08] |

Table S8: Profile of `em_program`: Rectangle vs. Vehicle program. Sensitivity (AUC drop) with refit-bootstrap 95% interval; bold: interval excludes zero.

| Descriptor | AUC | order | dur. coupl. | dur. marg. | rotation | jit. 0.25$u$ | jit. 0.5$u$ | jit. 1$u$ | jit. 2$u$ |
|---|---|---|---|---|---|---|---|---|---|
| Count | 0.561 | 0 (ex.) | 0 (ex.) | 0 (ex.) | 0 (ex.) | 0 (ex.) | 0 (ex.) | 0 (ex.) | 0 (ex.) |
| Dispersion | 0.957 | 0 (ex.) | 0 (ex.) | 0 (ex.) | **+0.139 [+0.08, +0.20]** | −0.001 [−0.01, +0.00] | −0.001 [−0.01, +0.01] | +0.001 [−0.01, +0.01] | +0.002 [−0.01, +0.02] |
| Classic | 0.539 | **−0.249 [−0.34, −0.16]** | 0 (ex.) | +0.032 [−0.08, +0.15] | 0 (ex.) | −0.000 [−0.02, +0.02] | +0.003 [−0.03, +0.04] | −0.004 [−0.06, +0.05] | +0.008 [−0.10, +0.11] |
| Dur. trend | 0.540 | +0.017 [−0.13, +0.16] | +0.038 [−0.12, +0.20] | +0.055 [−0.07, +0.18] | 0 (ex.) | 0 (ex.) | 0 (ex.) | 0 (ex.) | 0 (ex.) |
| RQA 0.5$u$ | 0.580 | +0.036 [−0.07, +0.14] | 0 (ex.) | 0 (ex.) | 0 (ex.) | +0.004 [−0.09, +0.09] | +0.061 [−0.08, +0.21] | +0.068 [−0.12, +0.25] | +0.000 [−0.17, +0.17] |
| RQA 1$u$ | 0.547 | +0.003 [−0.15, +0.16] | 0 (ex.) | 0 (ex.) | 0 (ex.) | −0.010 [−0.10, +0.08] | +0.003 [−0.13, +0.13] | −0.046 [−0.22, +0.13] | −0.067 [−0.26, +0.12] |
| Directed | 0.655 | −0.003 [−0.08, +0.07] | 0 (ex.) | 0 (ex.) | +0.090 [−0.00, +0.18] | +0.112 [−0.02, +0.24] | +0.141 [−0.01, +0.29] | +0.035 [−0.08, +0.15] | +0.044 [−0.10, +0.19] |
| FAG lattice | 0.658 | 0 (ex.) | −0.002 [−0.03, +0.03] | +0.001 [−0.03, +0.03] | +0.025 [−0.12, +0.17] | +0.022 [−0.04, +0.08] | −0.049 [−0.14, +0.04] | −0.032 [−0.17, +0.11] | +0.005 [−0.14, +0.15] |
| FAG AOI | 0.778 | 0 (ex.) | −0.005 [−0.02, +0.01] | −0.005 [−0.02, +0.01] | +0.079 [−0.04, +0.20] | −0.022 [−0.08, +0.04] | −0.013 [−0.10, +0.07] | −0.021 [−0.14, +0.09] | −0.044 [−0.14, +0.05] |
| FAG AOI pad. | 0.729 | 0 (ex.) | +0.001 [−0.02, +0.03] | +0.001 [−0.03, +0.03] | +0.056 [−0.06, +0.17] | −0.023 [−0.10, +0.06] | +0.017 [−0.09, +0.12] | +0.028 [−0.08, +0.13] | +0.049 [−0.09, +0.19] |

Table S9: Profile of `em_halves`: first vs. second half of a program reading. Sensitivity (AUC drop) with refit-bootstrap 95% interval; bold: interval excludes zero.

| Descriptor | AUC | order | dur. coupl. | dur. marg. | rotation | jit. 0.25$u$ | jit. 0.5$u$ | jit. 1$u$ | jit. 2$u$ |
|---|---|---|---|---|---|---|---|---|---|
| Count | 0.500 | 0 (ex.) | 0 (ex.) | 0 (ex.) | 0 (ex.) | 0 (ex.) | 0 (ex.) | 0 (ex.) | 0 (ex.) |
| Dispersion | 0.616 | 0 (ex.) | 0 (ex.) | 0 (ex.) | +0.012 [−0.05, +0.07] | −0.001 [−0.01, +0.01] | +0.000 [−0.01, +0.01] | +0.001 [−0.02, +0.02] | +0.008 [−0.02, +0.04] |
| Classic | 0.776 | **+0.172 [+0.11, +0.23]** | 0 (ex.) | +0.004 [−0.01, +0.02] | 0 (ex.) | −0.004 [−0.01, +0.00] | −0.003 [−0.02, +0.01] | −0.002 [−0.03, +0.02] | +0.043 [−0.00, +0.09] |
| Dur. trend | 0.582 | +0.048 [−0.04, +0.14] | +0.036 [−0.08, +0.15] | **+0.086 [+0.00, +0.17]** | 0 (ex.) | 0 (ex.) | 0 (ex.) | 0 (ex.) | 0 (ex.) |
| RQA 0.5$u$ | 0.613 | +0.058 [−0.04, +0.15] | 0 (ex.) | 0 (ex.) | 0 (ex.) | −0.036 [−0.09, +0.02] | −0.050 [−0.11, +0.01] | **−0.085 [−0.15, −0.02]** | −0.053 [−0.11, +0.01] |
| RQA 1$u$ | 0.680 | +0.081 [−0.00, +0.16] | 0 (ex.) | 0 (ex.) | 0 (ex.) | −0.006 [−0.02, +0.01] | −0.018 [−0.05, +0.01] | **−0.052 [−0.10, −0.01]** | −0.043 [−0.09, +0.01] |
| Directed | 0.634 | +0.033 [−0.04, +0.10] | 0 (ex.) | 0 (ex.) | −0.004 [−0.06, +0.05] | −0.010 [−0.07, +0.05] | −0.010 [−0.07, +0.05] | +0.002 [−0.07, +0.08] | +0.059 [−0.03, +0.15] |
| FAG lattice | 0.599 | 0 (ex.) | −0.008 [−0.03, +0.01] | −0.006 [−0.03, +0.01] | −0.010 [−0.07, +0.05] | −0.007 [−0.07, +0.05] | −0.000 [−0.08, +0.07] | +0.006 [−0.07, +0.08] | +0.036 [−0.08, +0.15] |
| FAG AOI | 0.682 | 0 (ex.) | −0.000 [−0.01, +0.01] | −0.002 [−0.01, +0.01] | **+0.098 [+0.03, +0.17]** | +0.036 [−0.03, +0.10] | +0.023 [−0.02, +0.07] | +0.045 [−0.03, +0.12] | **+0.113 [+0.01, +0.21]** |
| FAG AOI pad. | 0.669 | 0 (ex.) | −0.004 [−0.02, +0.02] | −0.001 [−0.02, +0.02] | +0.047 [−0.02, +0.11] | +0.027 [−0.04, +0.09] | +0.016 [−0.06, +0.09] | +0.016 [−0.05, +0.09] | +0.062 [−0.02, +0.14] |

Table S10: Profile of `em_expertise`: high vs. low/no self-rated programming expertise. Sensitivity (AUC drop) with refit-bootstrap 95% interval; bold: interval excludes zero.

| Descriptor | AUC | order | dur. coupl. | dur. marg. | rotation | jit. 0.25$u$ | jit. 0.5$u$ | jit. 1$u$ | jit. 2$u$ |
|---|---|---|---|---|---|---|---|---|---|
| Count | 0.472 | 0 (ex.) | 0 (ex.) | 0 (ex.) | 0 (ex.) | 0 (ex.) | 0 (ex.) | 0 (ex.) | 0 (ex.) |
| Dispersion | 0.469 | 0 (ex.) | 0 (ex.) | 0 (ex.) | −0.028 [−0.21, +0.15] | +0.006 [−0.03, +0.04] | +0.004 [−0.05, +0.06] | +0.007 [−0.06, +0.08] | −0.004 [−0.10, +0.09] |
| Classic | 0.438 | −0.032 [−0.28, +0.21] | 0 (ex.) | −0.023 [−0.34, +0.29] | 0 (ex.) | −0.002 [−0.04, +0.03] | −0.008 [−0.05, +0.03] | −0.043 [−0.10, +0.01] | −0.031 [−0.16, +0.09] |
| Dur. trend | 0.541 | +0.046 [−0.13, +0.23] | +0.070 [−0.23, +0.37] | +0.031 [−0.27, +0.33] | 0 (ex.) | 0 (ex.) | 0 (ex.) | 0 (ex.) | 0 (ex.) |
| RQA 0.5$u$ | 0.419 | −0.016 [−0.37, +0.34] | 0 (ex.) | 0 (ex.) | 0 (ex.) | −0.022 [−0.21, +0.17] | −0.033 [−0.11, +0.04] | +0.068 [−0.13, +0.27] | +0.056 [−0.21, +0.32] |
| RQA 1$u$ | 0.416 | −0.037 [−0.37, +0.30] | 0 (ex.) | 0 (ex.) | 0 (ex.) | −0.017 [−0.13, +0.09] | +0.016 [−0.24, +0.28] | +0.028 [−0.21, +0.27] | +0.090 [−0.24, +0.42] |
| Directed | 0.366 | +0.040 [−0.12, +0.20] | 0 (ex.) | 0 (ex.) | −0.018 [−0.10, +0.06] | +0.044 [−0.17, +0.26] | +0.013 [−0.25, +0.28] | −0.086 [−0.31, +0.14] | −0.083 [−0.32, +0.15] |
| FAG lattice | 0.447 | 0 (ex.) | +0.022 [−0.03, +0.08] | +0.028 [−0.07, +0.12] | −0.026 [−0.25, +0.20] | −0.035 [−0.28, +0.21] | −0.065 [−0.28, +0.15] | −0.062 [−0.39, +0.27] | −0.058 [−0.29, +0.18] |
| FAG AOI | 0.259 | 0 (ex.) | −0.009 [−0.07, +0.06] | −0.017 [−0.11, +0.07] | −0.231 [−0.48, +0.02] | −0.128 [−0.35, +0.09] | −0.143 [−0.37, +0.09] | −0.196 [−0.49, +0.10] | −0.110 [−0.39, +0.17] |
| FAG AOI pad. | 0.459 | 0 (ex.) | +0.015 [−0.05, +0.08] | +0.014 [−0.05, +0.07] | −0.034 [−0.26, +0.20] | +0.062 [−0.20, +0.33] | +0.010 [−0.25, +0.27] | −0.010 [−0.27, +0.25] | −0.023 [−0.30, +0.26] |

Table S11: Profile of `em_correct`: correct vs. incorrect answer, both programs pooled (confounded). Sensitivity (AUC drop) with refit-bootstrap 95% interval; bold: interval excludes zero.

| Descriptor | AUC | order | dur. coupl. | dur. marg. | rotation | jit. 0.25$u$ | jit. 0.5$u$ | jit. 1$u$ | jit. 2$u$ |
|---|---|---|---|---|---|---|---|---|---|
| Count | 0.495 | 0 (ex.) | 0 (ex.) | 0 (ex.) | 0 (ex.) | 0 (ex.) | 0 (ex.) | 0 (ex.) | 0 (ex.) |
| Dispersion | 0.716 | 0 (ex.) | 0 (ex.) | 0 (ex.) | **+0.096 [+0.01, +0.18]** | +0.000 [−0.01, +0.01] | +0.003 [−0.01, +0.01] | −0.001 [−0.01, +0.01] | +0.005 [−0.03, +0.04] |
| Classic | 0.501 | −0.123 [−0.27, +0.03] | 0 (ex.) | +0.054 [−0.07, +0.18] | 0 (ex.) | −0.000 [−0.02, +0.02] | −0.000 [−0.03, +0.03] | −0.004 [−0.06, +0.05] | −0.002 [−0.07, +0.07] |
| Dur. trend | 0.511 | −0.013 [−0.17, +0.15] | −0.023 [−0.17, +0.13] | +0.046 [−0.09, +0.18] | 0 (ex.) | 0 (ex.) | 0 (ex.) | 0 (ex.) | 0 (ex.) |
| RQA 0.5$u$ | 0.571 | +0.022 [−0.10, +0.14] | 0 (ex.) | 0 (ex.) | 0 (ex.) | +0.040 [−0.05, +0.13] | +0.056 [−0.05, +0.16] | +0.097 [−0.06, +0.25] | +0.061 [−0.12, +0.24] |
| RQA 1$u$ | 0.525 | −0.031 [−0.19, +0.13] | 0 (ex.) | 0 (ex.) | 0 (ex.) | +0.018 [−0.05, +0.09] | +0.069 [−0.04, +0.18] | +0.062 [−0.08, +0.20] | +0.009 [−0.18, +0.20] |
| Directed | 0.580 | −0.021 [−0.11, +0.07] | 0 (ex.) | 0 (ex.) | +0.043 [−0.05, +0.14] | +0.053 [−0.04, +0.15] | +0.103 [−0.02, +0.23] | +0.095 [−0.05, +0.24] | +0.058 [−0.12, +0.23] |
| FAG lattice | 0.664 | 0 (ex.) | +0.001 [−0.01, +0.02] | −0.002 [−0.02, +0.01] | +0.002 [−0.07, +0.08] | +0.002 [−0.07, +0.07] | −0.004 [−0.08, +0.08] | +0.051 [−0.07, +0.17] | +0.054 [−0.07, +0.18] |
| FAG AOI | 0.635 | 0 (ex.) | −0.004 [−0.03, +0.02] | −0.001 [−0.02, +0.02] | +0.076 [−0.09, +0.24] | +0.014 [−0.09, +0.12] | +0.052 [−0.06, +0.17] | −0.008 [−0.15, +0.13] | +0.031 [−0.11, +0.17] |
| FAG AOI pad. | 0.571 | 0 (ex.) | −0.006 [−0.03, +0.02] | −0.010 [−0.04, +0.02] | +0.052 [−0.09, +0.19] | −0.020 [−0.16, +0.12] | −0.022 [−0.16, +0.12] | −0.007 [−0.15, +0.14] | +0.004 [−0.15, +0.16] |

Table S12: Profile of `em_correctrect`: correct vs. incorrect answer, Rectangle program only. Sensitivity (AUC drop) with refit-bootstrap 95% interval; bold: interval excludes zero.

| Descriptor | AUC | order | dur. coupl. | dur. marg. | rotation | jit. 0.25$u$ | jit. 0.5$u$ | jit. 1$u$ | jit. 2$u$ |
|---|---|---|---|---|---|---|---|---|---|
| Count | 0.589 | 0 (ex.) | 0 (ex.) | 0 (ex.) | 0 (ex.) | 0 (ex.) | 0 (ex.) | 0 (ex.) | 0 (ex.) |
| Dispersion | 0.472 | 0 (ex.) | 0 (ex.) | 0 (ex.) | −0.062 [−0.25, +0.13] | +0.002 [−0.02, +0.02] | +0.006 [−0.03, +0.04] | +0.034 [−0.02, +0.09] | +0.073 [−0.02, +0.17] |
| Classic | 0.523 | −0.002 [−0.19, +0.19] | 0 (ex.) | +0.048 [−0.19, +0.28] | 0 (ex.) | +0.004 [−0.03, +0.04] | +0.008 [−0.05, +0.06] | −0.006 [−0.08, +0.06] | +0.003 [−0.08, +0.09] |
| Dur. trend | 0.507 | −0.069 [−0.23, +0.09] | −0.077 [−0.24, +0.08] | −0.037 [−0.29, +0.22] | 0 (ex.) | 0 (ex.) | 0 (ex.) | 0 (ex.) | 0 (ex.) |
| RQA 0.5$u$ | 0.451 | −0.054 [−0.38, +0.27] | 0 (ex.) | 0 (ex.) | 0 (ex.) | −0.046 [−0.25, +0.16] | −0.072 [−0.22, +0.08] | −0.004 [−0.26, +0.25] | −0.041 [−0.28, +0.20] |
| RQA 1$u$ | 0.462 | −0.065 [−0.38, +0.25] | 0 (ex.) | 0 (ex.) | 0 (ex.) | −0.015 [−0.15, +0.12] | −0.026 [−0.18, +0.13] | −0.046 [−0.24, +0.15] | −0.048 [−0.29, +0.20] |
| Directed | 0.533 | +0.006 [−0.22, +0.23] | 0 (ex.) | 0 (ex.) | −0.013 [−0.19, +0.17] | +0.000 [−0.18, +0.19] | +0.014 [−0.19, +0.22] | −0.000 [−0.22, +0.22] | +0.018 [−0.19, +0.23] |
| FAG lattice | 0.578 | 0 (ex.) | +0.007 [−0.05, +0.06] | +0.011 [−0.04, +0.06] | +0.021 [−0.11, +0.16] | +0.040 [−0.15, +0.23] | +0.025 [−0.11, +0.16] | +0.057 [−0.18, +0.29] | +0.097 [−0.06, +0.25] |
| FAG AOI | 0.658 | 0 (ex.) | +0.008 [−0.05, +0.07] | +0.005 [−0.06, +0.07] | +0.186 [−0.03, +0.40] | +0.092 [−0.16, +0.34] | **+0.167 [+0.01, +0.32]** | +0.144 [−0.08, +0.37] | +0.201 [−0.02, +0.42] |
| FAG AOI pad. | 0.544 | 0 (ex.) | +0.024 [−0.03, +0.08] | +0.020 [−0.05, +0.09] | +0.029 [−0.17, +0.23] | −0.027 [−0.19, +0.13] | −0.003 [−0.23, +0.22] | +0.069 [−0.18, +0.32] | +0.044 [−0.19, +0.28] |

Table S13: Profile of `em_correctveh`: correct vs. incorrect answer, Vehicle program only. Sensitivity (AUC drop) with refit-bootstrap 95% interval; bold: interval excludes zero.

| Descriptor | AUC | order | dur. coupl. | dur. marg. | rotation | jit. 0.25$u$ | jit. 0.5$u$ | jit. 1$u$ | jit. 2$u$ |
|---|---|---|---|---|---|---|---|---|---|
| Count | 0.444 | 0 (ex.) | 0 (ex.) | 0 (ex.) | 0 (ex.) | 0 (ex.) | 0 (ex.) | 0 (ex.) | 0 (ex.) |
| Dispersion | 0.476 | 0 (ex.) | 0 (ex.) | 0 (ex.) | +0.091 [−0.20, +0.38] | +0.006 [−0.02, +0.03] | +0.007 [−0.04, +0.06] | +0.015 [−0.04, +0.07] | +0.022 [−0.10, +0.15] |
| Classic | 0.500 | +0.070 [−0.15, +0.29] | 0 (ex.) | +0.025 [−0.20, +0.25] | 0 (ex.) | +0.000 [−0.04, +0.04] | +0.001 [−0.06, +0.06] | +0.003 [−0.05, +0.06] | +0.017 [−0.10, +0.14] |
| Dur. trend | 0.521 | +0.012 [−0.22, +0.24] | −0.024 [−0.26, +0.21] | +0.002 [−0.30, +0.31] | 0 (ex.) | 0 (ex.) | 0 (ex.) | 0 (ex.) | 0 (ex.) |
| RQA 0.5$u$ | 0.518 | −0.026 [−0.22, +0.17] | 0 (ex.) | 0 (ex.) | 0 (ex.) | +0.034 [−0.15, +0.21] | +0.026 [−0.23, +0.28] | +0.104 [−0.14, +0.35] | +0.098 [−0.11, +0.31] |
| RQA 1$u$ | 0.559 | +0.036 [−0.24, +0.31] | 0 (ex.) | 0 (ex.) | 0 (ex.) | −0.004 [−0.11, +0.10] | +0.093 [−0.10, +0.29] | +0.106 [−0.09, +0.30] | +0.140 [−0.07, +0.35] |
| Directed | 0.618 | +0.134 [−0.06, +0.33] | 0 (ex.) | 0 (ex.) | +0.080 [−0.14, +0.31] | +0.097 [−0.11, +0.30] | +0.120 [−0.05, +0.29] | +0.147 [−0.07, +0.37] | +0.088 [−0.15, +0.33] |
| FAG lattice | 0.574 | 0 (ex.) | +0.023 [−0.02, +0.07] | +0.021 [−0.05, +0.09] | +0.074 [−0.18, +0.33] | +0.066 [−0.13, +0.26] | +0.032 [−0.20, +0.26] | +0.069 [−0.17, +0.31] | +0.054 [−0.18, +0.29] |
| FAG AOI | 0.503 | 0 (ex.) | +0.000 [−0.06, +0.06] | −0.004 [−0.07, +0.06] | −0.010 [−0.29, +0.27] | −0.011 [−0.24, +0.22] | −0.017 [−0.30, +0.26] | +0.014 [−0.20, +0.23] | +0.001 [−0.26, +0.27] |
| FAG AOI pad. | 0.470 | 0 (ex.) | −0.007 [−0.05, +0.03] | −0.007 [−0.05, +0.04] | −0.148 [−0.43, +0.13] | +0.010 [−0.22, +0.24] | −0.052 [−0.26, +0.16] | +0.003 [−0.24, +0.24] | +0.015 [−0.24, +0.27] |

Table S14: Observed pooled AUC (all out-of-fold decision values ranked together; class-balanced participant folds), with the permutation $p$-value in brackets (200 permutations under the schemes of Table 2 of the article).

| Problem | Count | Dispersion | Classic | Dur. trend | RQA $0.5u$ | RQA $1u$ | Directed | FAG lattice | FAG AOI | FAG AOI pad. |
|---|---|---|---|---|---|---|---|---|---|---|
| `gb_tex_vd1` | 0.968 (.005) | 1.000 (.005) | 0.968 (.005) | 0.956 (.005) | 0.999 (.005) | 1.000 (.005) | 0.977 (.005) | 0.989 (.005) | – | – |
| `gb_vd1_blg` | 0.691 (.005) | 1.000 (.005) | 0.888 (.005) | 0.751 (.005) | 0.999 (.005) | 1.000 (.005) | 1.000 (.005) | 0.957 (.005) | – | – |
| `gb_vd1_vd2` | 0.633 (.005) | 0.703 (.005) | 0.636 (.005) | 0.791 (.005) | 0.672 (.005) | 0.715 (.005) | 0.700 (.005) | 0.666 (.005) | – | – |
| `gb_vd1_halves` | 0.500 (1.000) | 0.921 (.005) | 0.561 (.005) | 0.590 (.005) | 0.742 (.005) | 0.822 (.005) | 0.803 (.005) | 0.739 (.005) | – | – |
| `gb_tex_s1s2` | 0.516 (.065) | 0.517 (.224) | 0.537 (.005) | 0.545 (.015) | 0.574 (.005) | 0.515 (.159) | 0.531 (.070) | 0.588 (.005) | – | – |
| `sb_native` | 0.610 (.010) | 0.530 (.164) | 0.673 (.005) | 0.637 (.015) | 0.611 (.005) | 0.659 (.005) | 0.594 (.025) | 0.575 (.050) | – | – |
| `sb_compr` | 0.527 (.915) | 0.402 (.990) | 0.595 (.950) | 0.575 (.910) | 0.485 (.896) | 0.523 (.891) | 0.514 (.935) | 0.473 (.955) | – | – |
| `em_program` | 0.547 (.025) | 0.963 (.005) | 0.546 (.025) | 0.524 (.249) | 0.587 (.015) | 0.541 (.129) | 0.653 (.005) | 0.666 (.005) | 0.781 (.005) | 0.724 (.005) |
| `em_halves` | 0.500 (1.000) | 0.619 (.005) | 0.768 (.005) | 0.574 (.010) | 0.606 (.005) | 0.677 (.005) | 0.629 (.005) | 0.602 (.005) | 0.675 (.005) | 0.664 (.005) |
| `em_expertise` | 0.400 (.697) | 0.449 (.637) | 0.378 (.816) | 0.510 (.373) | 0.425 (.771) | 0.371 (.881) | 0.319 (.940) | 0.414 (.816) | 0.264 (.985) | 0.418 (.756) |
| `em_correct` | 0.463 (.741) | 0.710 (.478) | 0.497 (.522) | 0.505 (.393) | 0.554 (.179) | 0.518 (.328) | 0.559 (.279) | 0.659 (.015) | 0.638 (.254) | 0.569 (.746) |
| `em_correctrect` | 0.539 (.214) | 0.446 (.741) | 0.488 (.557) | 0.530 (.303) | 0.455 (.692) | 0.452 (.726) | 0.518 (.368) | 0.556 (.184) | 0.634 (.035) | 0.514 (.403) |
| `em_correctveh` | 0.434 (.701) | 0.445 (.731) | 0.458 (.672) | 0.474 (.627) | 0.490 (.507) | 0.564 (.159) | 0.588 (.119) | 0.536 (.299) | 0.465 (.592) | 0.465 (.627) |

Table S15: Range, over descriptors, of the mean of the permutation distribution of the observed AUC, under the within-fold AUC (1000 permutations) and the pooled AUC (200 permutations), both with class-balanced folds. On `em_correct` labels are permuted within programs and on `sb_compr` between the passages of a participant, so stimulus structure that survives the permutation raises the mean for descriptors that read it.

| Problem | Within-fold AUC | Pooled AUC |
|---|---|---|
| `gb_tex_vd1` | 0.499–0.501 | 0.491–0.496 |
| `gb_vd1_blg` | 0.499–0.502 | 0.494–0.497 |
| `gb_vd1_vd2` | 0.499–0.501 | 0.494–0.499 |
| `gb_vd1_halves` | 0.499–0.501 | 0.496–0.500 |
| `gb_tex_s1s2` | 0.499–0.501 | 0.494–0.498 |
| `sb_native` | 0.497–0.504 | 0.476–0.490 |
| `sb_compr` | 0.477–0.618 | 0.458–0.611 |
| `em_program` | 0.496–0.501 | 0.488–0.500 |
| `em_halves` | 0.498–0.502 | 0.487–0.500 |
| `em_expertise` | 0.489–0.504 | 0.466–0.492 |
| `em_correct` | 0.502–0.710 | 0.487–0.708 |
| `em_correctrect` | 0.496–0.505 | 0.476–0.492 |
| `em_correctveh` | 0.496–0.503 | 0.475–0.491 |

Table S16: Evidence ledger of the 89 interpreted cells. Primary: sensitivity with the within-fold AUC and class-balanced folds ($R = 6$, $B = 60$, seed 42), with its 95% refit-bootstrap interval (excluding zero by selection). Pooled AUC: the same cell with the pooled AUC. Partitions: range of the point estimate over 20 fold partitions. Long runs: range of the point estimate over three runs with $R = 20$ and $B = 300$ (seeds 42, 43, 44), and the number of those runs whose interval excluded zero. Trees: gradient-boosted trees ($R = 6$, $B = 60$, seed 42). Norm.: sensitivity divided by the observed AUC minus the mean of its permutation distribution, under the within-fold and the pooled AUC. Robust: sign and exclusion of zero hold in the primary run, the pooled run, every partition and every long run and, for the AOI graph, on the padded graph (Table S23 and Section 5.9 of the article). Bold: interval excludes zero.

| Problem | Descriptor | Transformation | Primary | Pooled AUC | Partitions | Long runs | Trees | Norm. | Norm. (pooled) | Robust |
|---|---|---|---|---|---|---|---|---|---|---|
| gb_tex_vd1 | Classic | duration marginal | +0.035 [+0.021, +0.048] | **+0.035 [+0.022, +0.048]** | +0.033 to +0.036 | +0.034 to +0.034 (3/3) | **+0.045 [+0.025, +0.065]** | +0.07 | +0.07 | ✓ |
| gb_tex_vd1 | Classic | order | −0.022 [−0.034, −0.010] | **−0.022 [−0.034, −0.009]** | −0.024 to −0.020 | −0.022 to −0.022 (3/3) | **−0.019 [−0.033, −0.005]** | −0.05 | −0.05 | ✓ |
| gb_tex_vd1 | Classic | jitter 0.25$u$ | −0.003 [−0.005, −0.001] | **−0.003 [−0.005, −0.001]** | −0.003 to −0.002 | −0.003 to −0.003 (3/3) | −0.002 [−0.009, +0.004] | −0.01 | −0.01 | ✓ |
| gb_tex_vd1 | Classic | jitter 0.5$u$ | −0.006 [−0.009, −0.002] | **−0.006 [−0.008, −0.003]** | −0.007 to −0.005 | −0.006 to −0.005 (3/3) | −0.005 [−0.011, +0.001] | −0.01 | −0.01 | ✓ |
| gb_tex_vd1 | Classic | jitter 1$u$ | −0.007 [−0.011, −0.003] | **−0.007 [−0.011, −0.003]** | −0.009 to −0.007 | −0.007 to −0.007 (3/3) | −0.006 [−0.014, +0.003] | −0.01 | −0.02 | ✓ |
| gb_tex_vd1 | Directed | order | −0.014 [−0.020, −0.008] | **−0.014 [−0.020, −0.008]** | −0.014 to −0.012 | −0.015 to −0.014 (3/3) | −0.004 [−0.011, +0.002] | −0.03 | −0.03 | ✓ |
| gb_tex_vd1 | Directed | jitter 0.25$u$ | −0.011 [−0.018, −0.004] | **−0.011 [−0.016, −0.005]** | −0.013 to −0.011 | −0.012 to −0.011 (3/3) | −0.004 [−0.010, +0.002] | −0.02 | −0.02 | ✓ |
| gb_tex_vd1 | Directed | jitter 0.5$u$ | −0.015 [−0.021, −0.009] | **−0.015 [−0.020, −0.009]** | −0.015 to −0.013 | −0.016 to −0.015 (3/3) | −0.005 [−0.011, +0.001] | −0.03 | −0.03 | ✓ |
| gb_tex_vd1 | Directed | jitter 1$u$ | −0.016 [−0.022, −0.011] | **−0.016 [−0.022, −0.010]** | −0.017 to −0.014 | −0.017 to −0.016 (3/3) | −0.006 [−0.012, +0.000] | −0.03 | −0.03 | ✓ |
| gb_tex_vd1 | Directed | jitter 2$u$ | −0.013 [−0.021, −0.005] | **−0.012 [−0.019, −0.005]** | −0.013 to −0.010 | −0.013 to −0.012 (3/3) | −0.002 [−0.011, +0.007] | −0.03 | −0.03 | ✓ |
| gb_tex_vd1 | Directed | rotation | −0.007 [−0.013, −0.002] | **−0.008 [−0.012, −0.003]** | −0.009 to −0.007 | −0.007 to −0.007 (3/3) | −0.002 [−0.009, +0.005] | −0.02 | −0.02 | ✓ |
| gb_tex_vd1 | Dispersion | rotation | +0.021 [+0.014, +0.029] | **+0.021 [+0.014, +0.029]** | +0.018 to +0.020 | +0.021 to +0.021 (3/3) | **+0.016 [+0.008, +0.024]** | +0.04 | +0.04 | ✓ |
| gb_tex_vd1 | Dur. trend | duration marginal | +0.473 [+0.442, +0.503] | **+0.474 [+0.446, +0.503]** | +0.466 to +0.479 | +0.454 to +0.469 (3/3) | **+0.371 [+0.318, +0.424]** | +1.03 | +1.03 | ✓ |
| gb_tex_vd1 | FAG lattice | jitter 1$u$ | +0.009 [+0.002, +0.016] | **+0.010 [+0.002, +0.017]** | +0.007 to +0.009 | +0.007 to +0.009 (3/3) | +0.005 [−0.003, +0.013] | +0.02 | +0.02 | ✓ |
| gb_tex_vd1 | FAG lattice | jitter 2$u$ | +0.049 [+0.037, +0.061] | **+0.050 [+0.038, +0.062]** | +0.044 to +0.047 | +0.047 to +0.048 (3/3) | **+0.041 [+0.024, +0.058]** | +0.10 | +0.10 | ✓ |
| gb_tex_vd1 | RQA 0.5$u$ | order | +0.010 [+0.003, +0.018] | **+0.011 [+0.004, +0.018]** | +0.009 to +0.011 | +0.010 to +0.010 (3/3) | **+0.013 [+0.004, +0.022]** | +0.02 | +0.02 | ✓ |
| gb_tex_vd1 | RQA 0.5$u$ | jitter 1$u$ | +0.004 [+0.001, +0.008] | **+0.004 [+0.001, +0.008]** | +0.003 to +0.004 | +0.003 to +0.004 (3/3) | +0.004 [−0.000, +0.007] | +0.01 | +0.01 | ✓ |
| gb_tex_vd1 | RQA 0.5$u$ | jitter 2$u$ | +0.015 [+0.009, +0.021] | **+0.016 [+0.010, +0.021]** | +0.017 to +0.019 | +0.017 to +0.018 (3/3) | **+0.014 [+0.009, +0.020]** | +0.03 | +0.03 | ✓ |
| gb_tex_vd1 | RQA 1$u$ | order | +0.006 [+0.001, +0.012] | **+0.007 [+0.001, +0.012]** | +0.006 to +0.007 | +0.006 to +0.006 (3/3) | **+0.007 [+0.001, +0.013]** | +0.01 | +0.01 | ✓ |
| gb_tex_vd1 | RQA 1$u$ | jitter 2$u$ | +0.006 [+0.001, +0.011] | **+0.007 [+0.002, +0.011]** | +0.006 to +0.007 | +0.006 to +0.006 (3/3) | **+0.007 [+0.001, +0.012]** | +0.01 | +0.01 | ✓ |
| gb_vd1_blg | Classic | duration marginal | +0.033 [+0.020, +0.046] | **+0.034 [+0.022, +0.047]** | +0.030 to +0.036 | +0.033 to +0.033 (3/3) | **+0.037 [+0.009, +0.065]** | +0.09 | +0.09 | ✓ |
| gb_vd1_blg | Classic | order | −0.114 [−0.137, −0.091] | **−0.112 [−0.133, −0.091]** | −0.114 to −0.110 | −0.114 to −0.114 (3/3) | **−0.118 [−0.148, −0.088]** | −0.30 | −0.29 | ✓ |
| gb_vd1_blg | Classic | jitter 0.25$u$ | −0.003 [−0.004, −0.001] | **−0.002 [−0.004, −0.001]** | −0.003 to −0.002 | −0.003 to −0.003 (3/3) | +0.001 [−0.008, +0.009] | −0.01 | −0.01 | ✓ |
| gb_vd1_blg | Classic | jitter 0.5$u$ | −0.007 [−0.010, −0.005] | **−0.007 [−0.009, −0.005]** | −0.007 to −0.006 | −0.007 to −0.007 (3/3) | −0.003 [−0.012, +0.006] | −0.02 | −0.02 | ✓ |
| gb_vd1_blg | Classic | jitter 1$u$ | −0.014 [−0.021, −0.007] | **−0.014 [−0.020, −0.008]** | −0.015 to −0.013 | −0.014 to −0.014 (3/3) | −0.009 [−0.021, +0.003] | −0.04 | −0.04 | ✓ |
| gb_vd1_blg | Classic | jitter 2$u$ | −0.014 [−0.023, −0.005] | **−0.013 [−0.021, −0.006]** | −0.015 to −0.013 | −0.016 to −0.014 (3/3) | −0.005 [−0.023, +0.012] | −0.04 | −0.03 | ✓ |
| gb_vd1_blg | Directed | order | +0.005 [+0.001, +0.008] | **+0.005 [+0.002, +0.008]** | +0.004 to +0.005 | +0.005 to +0.005 (3/3) | **+0.006 [+0.002, +0.010]** | +0.01 | +0.01 | ✓ |
| gb_vd1_blg | Directed | jitter 2$u$ | +0.002 [+0.000, +0.004] | **+0.002 [+0.001, +0.004]** | +0.002 to +0.002 | +0.002 to +0.002 (2/3) | +0.003 [−0.000, +0.006] | +0.00 | +0.00 | – |
| gb_vd1_blg | Dur. trend | duration marginal | +0.252 [+0.199, +0.304] | **+0.257 [+0.206, +0.308]** | +0.251 to +0.265 | +0.243 to +0.259 (3/3) | **+0.226 [+0.161, +0.291]** | +1.01 | +1.00 | ✓ |
| gb_vd1_blg | FAG lattice | duration marginal | −0.008 [−0.013, −0.004] | **−0.009 [−0.013, −0.005]** | −0.009 to −0.008 | −0.009 to −0.008 (3/3) | −0.004 [−0.009, +0.002] | −0.02 | −0.02 | ✓ |
| gb_vd1_blg | FAG lattice | jitter 1$u$ | +0.019 [+0.003, +0.035] | **+0.020 [+0.004, +0.035]** | +0.017 to +0.021 | +0.017 to +0.019 (3/3) | +0.011 [−0.000, +0.022] | +0.04 | +0.04 | ✓ |
| gb_vd1_blg | FAG lattice | jitter 2$u$ | +0.027 [+0.010, +0.044] | **+0.027 [+0.011, +0.043]** | +0.028 to +0.030 | +0.026 to +0.027 (3/3) | **+0.024 [+0.009, +0.039]** | +0.06 | +0.06 | ✓ |
| gb_vd1_blg | FAG lattice | rotation | +0.029 [+0.016, +0.042] | **+0.029 [+0.018, +0.041]** | +0.024 to +0.028 | +0.024 to +0.028 (3/3) | **+0.023 [+0.009, +0.036]** | +0.06 | +0.06 | ✓ |
| gb_vd1_blg | RQA 0.5$u$ | order | +0.013 [+0.006, +0.020] | **+0.013 [+0.007, +0.019]** | +0.012 to +0.014 | +0.013 to +0.013 (3/3) | **+0.014 [+0.005, +0.023]** | +0.03 | +0.03 | ✓ |
| gb_vd1_blg | RQA 0.5$u$ | jitter 2$u$ | +0.003 [+0.001, +0.006] | **+0.003 [+0.001, +0.006]** | +0.005 to +0.006 | +0.004 to +0.004 (3/3) | **+0.006 [+0.001, +0.010]** | +0.01 | +0.01 | ✓ |
| gb_vd1_blg | RQA 1$u$ | order | +0.001 [+0.000, +0.002] | **+0.001 [+0.000, +0.002]** | +0.001 to +0.001 | +0.001 to +0.001 (3/3) | +0.002 [−0.001, +0.005] | +0.00 | +0.00 | ✓ |
| gb_vd1_vd2 | Classic | duration marginal | +0.124 [+0.099, +0.150] | **+0.124 [+0.103, +0.144]** | +0.124 to +0.135 | +0.122 to +0.127 (3/3) | **+0.130 [+0.073, +0.187]** | +0.88 | +0.90 | ✓ |
| gb_vd1_vd2 | Classic | jitter 0.5$u$ | +0.004 [+0.000, +0.007] | +0.002 [−0.000, +0.004] | +0.001 to +0.003 | +0.003 to +0.003 (2/3) | +0.002 [−0.026, +0.031] | +0.03 | +0.01 | – |
| gb_vd1_vd2 | Directed | order | +0.040 [+0.015, +0.065] | **+0.042 [+0.018, +0.065]** | +0.038 to +0.045 | +0.037 to +0.039 (3/3) | **+0.059 [+0.007, +0.111]** | +0.20 | +0.21 | ✓ |
| gb_vd1_vd2 | Directed | jitter 1$u$ | +0.042 [+0.009, +0.075] | **+0.044 [+0.012, +0.076]** | +0.040 to +0.049 | +0.040 to +0.042 (3/3) | +0.048 [−0.000, +0.096] | +0.21 | +0.22 | ✓ |
| gb_vd1_vd2 | Directed | jitter 2$u$ | +0.069 [+0.044, +0.095] | **+0.071 [+0.045, +0.097]** | +0.065 to +0.072 | +0.067 to +0.068 (3/3) | **+0.086 [+0.040, +0.133]** | +0.35 | +0.35 | ✓ |
| gb_vd1_vd2 | Dispersion | jitter 0.5$u$ | +0.012 [+0.000, +0.023] | **+0.011 [+0.002, +0.021]** | +0.010 to +0.013 | +0.012 to +0.013 (3/3) | +0.035 [−0.015, +0.086] | +0.06 | +0.06 | ✓ |
| gb_vd1_vd2 | Dispersion | jitter 1$u$ | +0.045 [+0.026, +0.063] | **+0.043 [+0.027, +0.060]** | +0.039 to +0.043 | +0.040 to +0.048 (3/3) | **+0.069 [+0.005, +0.134]** | +0.21 | +0.21 | ✓ |
| gb_vd1_vd2 | Dispersion | jitter 2$u$ | +0.116 [+0.085, +0.148] | **+0.113 [+0.082, +0.143]** | +0.117 to +0.123 | +0.113 to +0.120 (3/3) | **+0.102 [+0.038, +0.166]** | +0.56 | +0.55 | ✓ |
| gb_vd1_vd2 | Dispersion | rotation | +0.098 [+0.068, +0.128] | **+0.096 [+0.069, +0.124]** | +0.108 to +0.114 | +0.100 to +0.103 (3/3) | **+0.108 [+0.045, +0.170]** | +0.47 | +0.47 | ✓ |
| gb_vd1_vd2 | Dur. trend | duration coupling | +0.158 [+0.131, +0.185] | **+0.162 [+0.140, +0.184]** | +0.154 to +0.162 | +0.157 to +0.158 (3/3) | **+0.176 [+0.133, +0.219]** | +0.54 | +0.55 | ✓ |
| gb_vd1_vd2 | Dur. trend | duration marginal | +0.286 [+0.227, +0.345] | **+0.292 [+0.233, +0.351]** | +0.280 to +0.294 | +0.289 to +0.297 (3/3) | **+0.272 [+0.216, +0.329]** | +0.98 | +0.98 | ✓ |
| gb_vd1_vd2 | Dur. trend | order | +0.157 [+0.127, +0.187] | **+0.161 [+0.134, +0.188]** | +0.150 to +0.160 | +0.157 to +0.157 (3/3) | **+0.178 [+0.133, +0.224]** | +0.54 | +0.54 | ✓ |
| gb_vd1_vd2 | FAG lattice | duration coupling | −0.015 [−0.029, −0.001] | −0.012 [−0.028, +0.003] | −0.017 to −0.012 | −0.017 to −0.015 (1/3) | −0.016 [−0.046, +0.013] | −0.09 | −0.07 | – |
| gb_vd1_vd2 | FAG lattice | duration marginal | −0.020 [−0.031, −0.009] | **−0.019 [−0.032, −0.007]** | −0.027 to −0.020 | −0.023 to −0.020 (3/3) | −0.014 [−0.038, +0.011] | −0.12 | −0.11 | ✓ |
| gb_vd1_vd2 | FAG lattice | jitter 2$u$ | +0.055 [+0.013, +0.096] | **+0.058 [+0.017, +0.099]** | +0.045 to +0.053 | +0.046 to +0.052 (3/3) | **+0.100 [+0.043, +0.156]** | +0.33 | +0.34 | ✓ |
| gb_vd1_vd2 | RQA 0.5$u$ | order | +0.035 [+0.011, +0.060] | **+0.034 [+0.013, +0.054]** | +0.032 to +0.040 | +0.033 to +0.035 (3/3) | +0.044 [−0.004, +0.092] | +0.20 | +0.19 | ✓ |

| Problem | Descriptor | Transformation | Primary | Pooled AUC | Partitions | Long runs | Trees | Norm. | Norm. (pooled) | Robust |
|---|---|---|---|---|---|---|---|---|---|---|
| gb_vd1_vd2 | RQA 0.5$u$ | jitter 1$u$ | +0.046 [+0.003, +0.090] | **+0.044 [+0.000, +0.087]** | +0.033 to +0.040 | +0.035 to +0.039 (3/3) | +0.049 [−0.023, +0.121] | +0.26 | +0.25 | ✓ |
| gb_vd1_vd2 | RQA 0.5$u$ | jitter 2$u$ | +0.108 [+0.071, +0.145] | **+0.106 [+0.071, +0.140]** | +0.107 to +0.114 | +0.106 to +0.115 (3/3) | **+0.106 [+0.041, +0.170]** | +0.61 | +0.61 | ✓ |
| gb_vd1_vd2 | RQA 1$u$ | order | +0.073 [+0.043, +0.103] | **+0.073 [+0.045, +0.101]** | +0.069 to +0.076 | +0.072 to +0.073 (3/3) | **+0.086 [+0.032, +0.140]** | +0.33 | +0.33 | ✓ |
| gb_vd1_vd2 | RQA 1$u$ | jitter 1$u$ | +0.043 [+0.007, +0.078] | **+0.041 [+0.007, +0.075]** | +0.041 to +0.048 | +0.040 to +0.042 (3/3) | +0.044 [−0.005, +0.093] | +0.19 | +0.19 | ✓ |
| gb_vd1_vd2 | RQA 1$u$ | jitter 2$u$ | +0.118 [+0.080, +0.156] | **+0.115 [+0.077, +0.152]** | +0.108 to +0.115 | +0.110 to +0.119 (3/3) | **+0.125 [+0.054, +0.195]** | +0.54 | +0.52 | ✓ |
| gb_vd1_halves | Classic | order | −0.090 [−0.110, −0.069] | **−0.091 [−0.111, −0.071]** | −0.098 to −0.088 | −0.089 to −0.087 (3/3) | −0.055 [−0.114, +0.003] | −1.40 | −1.45 | ✓ |
| gb_vd1_halves | Directed | order | +0.036 [+0.017, +0.054] | **+0.036 [+0.019, +0.052]** | +0.032 to +0.036 | +0.037 to +0.037 (3/3) | +0.027 [−0.003, +0.058] | +0.12 | +0.12 | ✓ |
| gb_vd1_halves | Directed | jitter 0.5$u$ | +0.033 [+0.014, +0.052] | **+0.031 [+0.011, +0.052]** | +0.026 to +0.031 | +0.031 to +0.033 (3/3) | +0.029 [−0.007, +0.065] | +0.11 | +0.10 | ✓ |
| gb_vd1_halves | Directed | jitter 1$u$ | +0.090 [+0.060, +0.121] | **+0.089 [+0.061, +0.117]** | +0.087 to +0.093 | +0.087 to +0.093 (3/3) | **+0.102 [+0.060, +0.145]** | +0.29 | +0.29 | ✓ |
| gb_vd1_halves | Directed | jitter 2$u$ | +0.191 [+0.161, +0.220] | **+0.190 [+0.162, +0.218]** | +0.167 to +0.172 | +0.186 to +0.187 (3/3) | **+0.192 [+0.140, +0.243]** | +0.62 | +0.62 | ✓ |
| gb_vd1_halves | Dispersion | jitter 0.5$u$ | +0.007 [+0.002, +0.012] | **+0.007 [+0.002, +0.011]** | +0.006 to +0.008 | +0.007 to +0.007 (3/3) | +0.001 [−0.011, +0.013] | +0.02 | +0.02 | ✓ |
| gb_vd1_halves | Dispersion | jitter 1$u$ | +0.029 [+0.018, +0.041] | **+0.029 [+0.020, +0.038]** | +0.031 to +0.033 | +0.030 to +0.032 (3/3) | **+0.028 [+0.010, +0.046]** | +0.07 | +0.07 | ✓ |
| gb_vd1_halves | Dispersion | jitter 2$u$ | +0.130 [+0.111, +0.150] | **+0.129 [+0.110, +0.149]** | +0.123 to +0.126 | +0.122 to +0.129 (3/3) | **+0.123 [+0.097, +0.150]** | +0.31 | +0.31 | ✓ |
| gb_vd1_halves | Dispersion | rotation | +0.249 [+0.224, +0.275] | **+0.247 [+0.223, +0.271]** | +0.249 to +0.253 | +0.246 to +0.249 (3/3) | **+0.177 [+0.139, +0.215]** | +0.59 | +0.58 | ✓ |
| gb_vd1_halves | Dur. trend | duration coupling | +0.085 [+0.040, +0.130] | **+0.086 [+0.042, +0.129]** | +0.078 to +0.090 | +0.078 to +0.082 (3/3) | +0.060 [−0.002, +0.121] | +0.96 | +0.91 | ✓ |
| gb_vd1_halves | Dur. trend | duration marginal | +0.082 [+0.044, +0.120] | **+0.088 [+0.051, +0.125]** | +0.077 to +0.085 | +0.089 to +0.095 (3/3) | +0.024 [−0.051, +0.098] | +0.93 | +0.94 | ✓ |
| gb_vd1_halves | Dur. trend | order | +0.076 [+0.030, +0.122] | **+0.079 [+0.031, +0.126]** | +0.072 to +0.080 | +0.080 to +0.081 (3/3) | +0.033 [−0.045, +0.112] | +0.86 | +0.84 | ✓ |
| gb_vd1_halves | FAG lattice | jitter 2$u$ | +0.088 [+0.053, +0.123] | **+0.087 [+0.052, +0.121]** | +0.084 to +0.090 | +0.082 to +0.082 (3/3) | **+0.111 [+0.049, +0.173]** | +0.37 | +0.36 | ✓ |
| gb_vd1_halves | RQA 0.5$u$ | order | +0.046 [+0.023, +0.069] | **+0.046 [+0.025, +0.068]** | +0.043 to +0.049 | +0.047 to +0.048 (3/3) | **+0.055 [+0.014, +0.096]** | +0.19 | +0.19 | ✓ |
| gb_vd1_halves | RQA 0.5$u$ | jitter 1$u$ | +0.079 [+0.051, +0.106] | **+0.079 [+0.051, +0.107]** | +0.068 to +0.074 | +0.070 to +0.071 (3/3) | **+0.080 [+0.031, +0.129]** | +0.32 | +0.33 | ✓ |
| gb_vd1_halves | RQA 0.5$u$ | jitter 2$u$ | +0.143 [+0.097, +0.189] | **+0.143 [+0.099, +0.188]** | +0.137 to +0.142 | +0.145 to +0.150 (3/3) | **+0.157 [+0.101, +0.214]** | +0.59 | +0.59 | ✓ |
| gb_vd1_halves | RQA 1$u$ | order | +0.088 [+0.063, +0.112] | **+0.089 [+0.066, +0.111]** | +0.084 to +0.090 | +0.085 to +0.086 (3/3) | **+0.113 [+0.077, +0.149]** | +0.27 | +0.27 | ✓ |
| gb_vd1_halves | RQA 1$u$ | jitter 0.5$u$ | +0.026 [+0.006, +0.046] | **+0.026 [+0.007, +0.045]** | +0.018 to +0.021 | +0.028 to +0.031 (3/3) | +0.025 [−0.011, +0.061] | +0.08 | +0.08 | ✓ |
| gb_vd1_halves | RQA 1$u$ | jitter 1$u$ | +0.088 [+0.054, +0.121] | **+0.089 [+0.059, +0.118]** | +0.086 to +0.090 | +0.089 to +0.092 (3/3) | **+0.106 [+0.071, +0.142]** | +0.27 | +0.27 | ✓ |
| gb_vd1_halves | RQA 1$u$ | jitter 2$u$ | +0.182 [+0.151, +0.214] | **+0.183 [+0.154, +0.213]** | +0.170 to +0.174 | +0.179 to +0.183 (3/3) | **+0.212 [+0.172, +0.253]** | +0.56 | +0.57 | ✓ |
| gb_tex_s1s2 | RQA 0.5$u$ | order | +0.093 [+0.013, +0.174] | **+0.095 [+0.017, +0.173]** | +0.077 to +0.097 | +0.064 to +0.083 (1/3) | +0.066 [−0.037, +0.168] | +1.23 | +1.19 | – |
| gb_tex_s1s2 | RQA 0.5$u$ | jitter 0.5$u$ | +0.071 [+0.020, +0.123] | **+0.071 [+0.017, +0.125]** | +0.048 to +0.062 | +0.057 to +0.065 (0/3) | +0.035 [−0.065, +0.135] | +0.94 | +0.89 | – |
| sb_native | RQA 0.5$u$ | jitter 2$u$ | +0.088 [+0.012, +0.164] | **+0.096 [+0.009, +0.182]** | +0.104 to +0.125 | +0.103 to +0.117 (3/3) | −0.002 [−0.084, +0.081] | +0.79 | +0.77 | ✓ |
| sb_native | RQA 0.5$u$ | jitter 1$u$ | +0.079 [+0.007, +0.150] | +0.085 [−0.005, +0.174] | +0.055 to +0.072 | +0.069 to +0.076 (2/3) | −0.020 [−0.102, +0.062] | +0.71 | +0.68 | – |
| sb_native | RQA 1$u$ | jitter 2$u$ | +0.140 [+0.045, +0.235] | **+0.141 [+0.035, +0.246]** | +0.139 to +0.171 | +0.154 to +0.155 (3/3) | **+0.100 [+0.009, +0.190]** | +0.85 | +0.81 | ✓ |
| sb_native | RQA 1$u$ | jitter 1$u$ | +0.086 [+0.016, +0.157] | **+0.085 [+0.009, +0.161]** | +0.068 to +0.087 | +0.080 to +0.084 (3/3) | +0.071 [−0.003, +0.145] | +0.52 | +0.49 | ✓ |
| em_program | Dispersion | rotation | +0.139 [+0.078, +0.199] | **+0.160 [+0.108, +0.213]** | +0.137 to +0.165 | +0.134 to +0.142 (3/3) | **+0.166 [+0.088, +0.244]** | +0.30 | +0.34 | ✓ |
| em_halves | Classic | order | +0.172 [+0.112, +0.231] | **+0.171 [+0.120, +0.222]** | +0.164 to +0.191 | +0.171 to +0.174 (3/3) | **+0.128 [+0.033, +0.222]** | +0.62 | +0.62 | ✓ |
| em_halves | FAG AOI | jitter 2$u$ | +0.113 [+0.014, +0.212] | **+0.111 [+0.023, +0.198]** | +0.101 to +0.122 | +0.111 to +0.115 (3/3) | +0.115 [−0.007, +0.237] | +0.62 | +0.61 | – |
| em_halves | FAG AOI | rotation | +0.098 [+0.031, +0.166] | **+0.093 [+0.028, +0.158]** | +0.081 to +0.100 | +0.102 to +0.114 (3/3) | +0.058 [−0.060, +0.176] | +0.54 | +0.51 | – |
| em_halves | RQA 0.5$u$ | jitter 1$u$ | −0.085 [−0.152, −0.018] | **−0.085 [−0.145, −0.025]** | −0.089 to −0.073 | −0.085 to −0.078 (3/3) | −0.089 [−0.189, +0.011] | −0.77 | −0.75 | ✓ |
| em_halves | RQA 1$u$ | jitter 1$u$ | −0.052 [−0.098, −0.006] | **−0.052 [−0.086, −0.017]** | −0.063 to −0.045 | −0.050 to −0.045 (2/3) | −0.053 [−0.127, +0.020] | −0.29 | −0.28 | – |

Table S19: Extended simulation designs ($R = 6$, $B = 60$; 40 datasets per row). Target: the reference value (60 datasets with disjoint seeds, $R = 20$) with its Monte Carlo standard error, or zero by exchangeability. Unequal: one to four trials per class and participant (W) or per participant (B); 1:3 and 16:49: class imbalance; heterogeneous: participant offset SD 100 px and participant-specific scaling.

| Design | Size | Contrast | Target $\theta$ (SE) | Coverage | Excl. 0 | Width | Obs. AUC |
|---|---|---|---|---|---|---|---|
| order hides spread (classic) | 65 × 2 (B) | duration marginal | +0.006 (0.009) | 0.975 | 0.025 | 0.397 | 0.520 |
| order hides spread (classic) | 65 × 2 (B) | order | −0.499 (0.010) | 1.000 | 1.000 | 0.405 | 0.520 |
| order hides spread (classic) | 60 × 4 (W) | duration marginal | −0.001 (0.005) | 1.000 | 0.000 | 0.296 | 0.509 |
| order hides spread (classic) | 60 × 4 (W) | order | −0.497 (0.008) | 1.000 | 1.000 | 0.292 | 0.509 |
| no signal (rqa) | 65 × 2, heterog. (B) | order | 0 (by arg.) | 1.000 | 0.000 | 0.429 | 0.493 |
| no signal (rqa) | 65 × 2, heterog. (B) | jitter 160 px | +0.001 (0.012) | 1.000 | 0.000 | 0.498 | 0.493 |
| no signal (rqa) | 65 × 2, heterog. (B) | jitter 40 px | +0.005 (0.011) | 1.000 | 0.000 | 0.451 | 0.493 |
| no signal (rqa) | 65, 16:49 (B) | order | 0 (by arg.) | 0.925 | 0.075 | 0.502 | 0.474 |
| no signal (rqa) | 65, 16:49 (B) | jitter 160 px | +0.005 (0.012) | 1.000 | 0.000 | 0.605 | 0.474 |
| no signal (rqa) | 65, 16:49 (B) | jitter 40 px | +0.005 (0.012) | 1.000 | 0.000 | 0.573 | 0.474 |
| no signal (rqa) | 65, unequal (B) | order | 0 (by arg.) | 1.000 | 0.000 | 0.389 | 0.508 |
| no signal (rqa) | 65, unequal (B) | jitter 160 px | +0.014 (0.008) | 1.000 | 0.000 | 0.455 | 0.508 |

| Design | Size | Contrast | Target $\theta$ (SE) | Coverage | Excl. 0 | Width | Obs. AUC |
|---|---|---|---|---|---|---|---|
| no signal (rqa) | 65, unequal (B) | jitter 40 px | +0.016 (0.008) | 1.000 | 0.000 | 0.446 | 0.508 |
| no signal (rqa) | 60 × 4, heterog. (W) | order | 0 (by arg.) | 1.000 | 0.000 | 0.377 | 0.508 |
| no signal (rqa) | 60 × 4, heterog. (W) | jitter 160 px | −0.009 (0.006) | 1.000 | 0.000 | 0.375 | 0.508 |
| no signal (rqa) | 60 × 4, heterog. (W) | jitter 40 px | −0.007 (0.006) | 1.000 | 0.000 | 0.362 | 0.508 |
| no signal (rqa) | 60, 1:3 (W) | order | 0 (by arg.) | 1.000 | 0.000 | 0.393 | 0.510 |
| no signal (rqa) | 60, 1:3 (W) | jitter 160 px | +0.010 (0.009) | 1.000 | 0.000 | 0.445 | 0.510 |
| no signal (rqa) | 60, 1:3 (W) | jitter 40 px | +0.007 (0.009) | 1.000 | 0.000 | 0.432 | 0.510 |
| no signal (rqa) | 60, unequal (W) | order | 0 (by arg.) | 1.000 | 0.000 | 0.295 | 0.490 |
| no signal (rqa) | 60, unequal (W) | jitter 160 px | +0.012 (0.006) | 1.000 | 0.000 | 0.349 | 0.490 |
| no signal (rqa) | 60, unequal (W) | jitter 40 px | +0.008 (0.006) | 1.000 | 0.000 | 0.334 | 0.490 |
| weak order contrast (directed) | 65 × 2 (B) | order | +0.118 (0.008) | 1.000 | 0.075 | 0.424 | 0.632 |
| weak order contrast (directed) | 65 × 2 (B) | jitter 40 px | +0.015 (0.007) | 1.000 | 0.000 | 0.425 | 0.632 |
| weak order contrast (directed) | 65 × 2 (B) | rotation | +0.000 (0.006) | 1.000 | 0.000 | 0.373 | 0.632 |
| weak order contrast (directed) | 60 × 4 (W) | order | +0.128 (0.007) | 0.975 | 0.350 | 0.307 | 0.639 |
| weak order contrast (directed) | 60 × 4 (W) | jitter 40 px | +0.015 (0.005) | 1.000 | 0.000 | 0.298 | 0.639 |
| weak order contrast (directed) | 60 × 4 (W) | rotation | −0.004 (0.005) | 1.000 | 0.000 | 0.265 | 0.639 |
| weak spread contrast (rqa) | 65 × 2, heterog. (B) | order | 0 (by arg.) | 1.000 | 0.000 | 0.282 | 0.671 |
| weak spread contrast (rqa) | 65 × 2, heterog. (B) | jitter 40 px | +0.127 (0.007) | 1.000 | 0.125 | 0.412 | 0.671 |
| weak spread contrast (rqa) | 65, 16:49 (B) | order | 0 (by arg.) | 1.000 | 0.000 | 0.172 | 0.871 |
| weak spread contrast (rqa) | 65, 16:49 (B) | jitter 40 px | +0.315 (0.005) | 1.000 | 0.975 | 0.394 | 0.871 |
| weak spread contrast (rqa) | 65, unequal (B) | order | 0 (by arg.) | 1.000 | 0.000 | 0.106 | 0.878 |
| weak spread contrast (rqa) | 65, unequal (B) | jitter 40 px | +0.310 (0.004) | 1.000 | 1.000 | 0.302 | 0.878 |
| weak spread contrast (rqa) | 65 × 2 (B) | order | 0 (by arg.) | 1.000 | 0.000 | 0.133 | 0.882 |
| weak spread contrast (rqa) | 65 × 2 (B) | jitter 40 px | +0.306 (0.004) | 1.000 | 1.000 | 0.334 | 0.882 |
| weak spread contrast (rqa) | 60 × 4, heterog. (W) | order | 0 (by arg.) | 1.000 | 0.000 | 0.152 | 0.693 |
| weak spread contrast (rqa) | 60 × 4, heterog. (W) | jitter 40 px | +0.143 (0.003) | 1.000 | 0.600 | 0.250 | 0.693 |
| weak spread contrast (rqa) | 60, 1:3 (W) | order | 0 (by arg.) | 1.000 | 0.000 | 0.093 | 0.872 |
| weak spread contrast (rqa) | 60, 1:3 (W) | jitter 40 px | +0.302 (0.004) | 0.975 | 1.000 | 0.290 | 0.872 |
| weak spread contrast (rqa) | 60, unequal (W) | order | 0 (by arg.) | 1.000 | 0.000 | 0.059 | 0.881 |
| weak spread contrast (rqa) | 60, unequal (W) | jitter 40 px | +0.300 (0.003) | 1.000 | 1.000 | 0.209 | 0.881 |
| weak spread contrast (rqa) | 60 × 4 (W) | order | 0 (by arg.) | 1.000 | 0.000 | 0.070 | 0.874 |
| weak spread contrast (rqa) | 60 × 4 (W) | jitter 40 px | +0.302 (0.003) | 1.000 | 1.000 | 0.247 | 0.874 |
| signal elsewhere, no offset (rqa) | 65 × 2, heterog. (B) | order | 0 (by arg.) | 1.000 | 0.000 | 0.299 | 0.637 |
| signal elsewhere, no offset (rqa) | 65, 16:49 (B) | order | 0 (by arg.) | 1.000 | 0.000 | 0.213 | 0.824 |
| signal elsewhere, no offset (rqa) | 65, unequal (B) | order | 0 (by arg.) | 1.000 | 0.000 | 0.126 | 0.836 |
| signal elsewhere, no offset (rqa) | 60 × 4, heterog. (W) | order | 0 (by arg.) | 1.000 | 0.000 | 0.177 | 0.660 |
| signal elsewhere, no offset (rqa) | 60, 1:3 (W) | order | 0 (by arg.) | 1.000 | 0.000 | 0.113 | 0.833 |
| signal elsewhere, no offset (rqa) | 60, unequal (W) | order | 0 (by arg.) | 1.000 | 0.000 | 0.072 | 0.841 |

Table S17: Bootstrap standard error against the empirical spread of the estimate in the calibration campaign ($R = 6$, $B = 60$; 40 datasets per design and size). SE ratio: mean bootstrap SE divided by the SD of the estimate across datasets, per contrast; median and range over the contrasts of the cell. Coverage of the target by normal intervals at nominal 80%, 90% and 95%.

| Design | Size | Contrasts | SE ratio, median (range) | Coverage at 80% | at 90% | at 95% |
|---|---|---|---|---|---|---|
| no signal | 65 × 2 (B) | 8 | 1.66 (1.35–2.04) | 0.959 | 0.988 | 0.997 |
| no signal | 95 × 20 (B) | 8 | 1.48 (1.23–2.03) | 0.922 | 0.978 | 0.988 |
| no signal | 112 × 2 (W) | 8 | 1.80 (1.66–2.02) | 0.963 | 0.994 | 1.000 |
| no signal | 60 × 4 (W) | 8 | 1.80 (1.48–2.22) | 0.966 | 0.987 | 1.000 |
| signal elsewhere | 65 × 2 (B) | 5 | 2.40 (1.94–3.10) | 0.985 | 0.990 | 1.000 |
| signal elsewhere | 95 × 20 (B) | 5 | 1.46 (1.28–1.65) | 0.925 | 0.985 | 1.000 |
| signal elsewhere | 112 × 2 (W) | 5 | 2.35 (2.04–2.82) | 0.990 | 1.000 | 1.000 |
| signal elsewhere | 60 × 4 (W) | 5 | 2.65 (1.76–3.37) | 0.995 | 1.000 | 1.000 |
| signal elsewhere, no offset | 65 × 2 (B) | 3 | 2.40 (1.87–2.58) | 1.000 | 1.000 | 1.000 |
| signal elsewhere, no offset | 95 × 20 (B) | 3 | 1.29 (1.28–1.66) | 0.925 | 0.967 | 0.983 |
| signal elsewhere, no offset | 112 × 2 (W) | 3 | 1.97 (1.57–2.82) | 0.975 | 1.000 | 1.000 |
| signal elsewhere, no offset | 60 × 4 (W) | 3 | 2.19 (1.67–3.37) | 0.975 | 1.000 | 1.000 |

Table S18: Corrected refit bootstrap (Equation 1) against the direct refit bootstrap, which averages all $R = 6$ transformed replicates inside each of the $B = 60$ draws, on the same 200 datasets per cell. Coverage of the target, mean width, and the share of intervals excluding zero.

| | | | | Coverage | | Width | | Excl. 0 | |
|---|---|---|---|---|---|---|---|---|---|
| Design | Size | Descr. | Contrast | direct | corr. | direct | corr. | direct | corr. |
| no signal | 65 × 2 (B) | rqa | order | 0.985 | 0.990 | 0.367 | 0.434 | 0.015 | 0.010 |
| no signal | 65 × 2 (B) | rqa | jitter 160 px | 0.995 | 1.000 | 0.419 | 0.505 | 0.005 | 0.000 |
| no signal | 65 × 2 (B) | rqa | jitter 40 px | 0.985 | 0.990 | 0.417 | 0.498 | 0.015 | 0.010 |
| no signal | 60 × 4 (W) | classic | order | 0.995 | 0.985 | 0.241 | 0.286 | 0.005 | 0.015 |
| no signal | 60 × 4 (W) | rqa | order | 0.990 | 0.995 | 0.272 | 0.324 | 0.010 | 0.005 |
| no signal | 60 × 4 (W) | rqa | jitter 160 px | 0.995 | 1.000 | 0.313 | 0.379 | 0.005 | 0.005 |
| no signal | 60 × 4 (W) | rqa | jitter 40 px | 0.990 | 1.000 | 0.314 | 0.376 | 0.010 | 0.000 |
| signal elsewhere | 65 × 2 (B) | rqa | order | 1.000 | 1.000 | 0.125 | 0.161 | 0.000 | 0.000 |
| signal elsewhere | 65 × 2 (B) | rqa | jitter 40 px | 1.000 | 1.000 | 0.219 | 0.345 | 1.000 | 0.955 |
| signal elsewhere | 60 × 4 (W) | rqa | order | 1.000 | 1.000 | 0.070 | 0.089 | 0.000 | 0.000 |
| signal elsewhere | 60 × 4 (W) | rqa | jitter 40 px | 0.995 | 1.000 | 0.147 | 0.238 | 1.000 | 1.000 |
| signal elsewhere, no offset | 60 × 4 (W) | rqa | order | 1.000 | 1.000 | 0.070 | 0.089 | 0.000 | 0.000 |

Table S20: Controls for the EMIP program and correctness contrasts (within-fold AUC). Program contrast: AUC on real gaze and on synthetic gaze, in which every trial is replaced by the same number of fixations drawn uniformly over the bounding box of its program's code tokens. Correctness, both programs: AUC over all correct–incorrect pairs of a test fold, the mean of its permutation distribution when labels are permuted within programs, and the AUC over pairs from the same program and test fold with a refit participant-bootstrap interval (200 draws) and a one-sided permutation $p$-value (1000 permutations within programs). Descriptors with no spatial features are not affected by the synthetic control and are marked with a dash.

| | Program contrast | | Correctness, both programs | | |
|---|---|---|---|---|---|
| Descriptor | Real gaze | Synthetic gaze | All pairs | Perm. mean | Same-program pairs [95% interval] |
| Count | – | – | 0.495 | 0.515 | 0.517 [0.317, 0.646] ($p = 0.455$) |
| Dispersion | 0.957 | 1.000 | 0.716 | 0.710 | 0.490 [0.398, 0.581] ($p = 0.594$) |
| Classic | – | – | 0.501 | 0.505 | 0.523 [0.319, 0.625] ($p = 0.394$) |
| Dur. trend | – | – | 0.511 | 0.502 | 0.538 [0.373, 0.640] ($p = 0.262$) |
| RQA $0.5u$ | 0.580 | 0.604 | 0.571 | 0.517 | 0.527 [0.403, 0.645] ($p = 0.304$) |
| RQA $1u$ | 0.547 | 0.648 | 0.525 | 0.502 | 0.509 [0.384, 0.636] ($p = 0.462$) |
| Directed | 0.655 | 0.554 | 0.580 | 0.540 | 0.562 [0.410, 0.629] ($p = 0.130$) |
| FAG lattice | 0.658 | 0.859 | 0.664 | 0.566 | 0.577 [0.484, 0.685] ($p = 0.042$) |
| FAG AOI | 0.778 | 0.879 | 0.635 | 0.619 | 0.559 [0.441, 0.666] ($p = 0.087$) |
| FAG AOI pad. | 0.729 | 0.789 | 0.571 | 0.591 | 0.494 [0.345, 0.603] ($p = 0.575$) |

Table S21: EMIP expertise: AUC of each of the ten test folds and of the corresponding training set, for three descriptors (linear SVM; class-balanced folds).

| Fold | Classic | | Directed | | FAG AOI | |
|---|---|---|---|---|---|---|
| | test | train | test | train | test | train |
| 1 | 0.686 | 0.532 | 0.400 | 0.593 | 0.229 | 0.588 |
| 2 | 0.167 | 0.582 | 0.033 | 0.616 | 0.300 | 0.533 |
| 3 | 0.143 | 0.595 | 0.381 | 0.589 | 0.095 | 0.598 |
| 4 | 0.929 | 0.497 | 0.964 | 0.485 | 0.107 | 0.523 |
| 5 | 0.429 | 0.551 | 0.257 | 0.594 | 0.200 | 0.595 |
| 6 | 0.000 | 0.605 | 0.133 | 0.608 | 0.467 | 0.532 |
| 7 | 0.300 | 0.562 | 0.500 | 0.576 | 0.100 | 0.549 |
| 8 | 0.733 | 0.528 | 0.367 | 0.539 | 0.533 | 0.531 |
| 9 | 0.214 | 0.533 | 0.214 | 0.538 | 0.286 | 0.542 |
| 10 | 0.844 | 0.528 | 0.438 | 0.569 | 0.344 | 0.567 |
| Mean | 0.444 | 0.551 | 0.369 | 0.571 | 0.266 | 0.556 |

Table S22: Share of dwell time on the AOI triangulation, negative / positive class, for every EMIP problem and transformation (mean over trials of the mean over $R = 6$ replicates). Fixations off the triangulation are dropped by the AOI graph. On the lattice graph and on the padded AOI graph the share was at least 0.998 in every cell.

| Transformation | `em_program` | `em_halves` | `em_expertise` | `em_correct` | `em_correctrect` | `em_correctveh` |
|---|---|---|---|---|---|---|
| none (observed) | 0.974 / 0.975 | 0.960 / 0.990 | 0.977 / 0.973 | 0.973 / 0.974 | 0.972 / 0.972 | 0.978 / 0.975 |
| order | 0.974 / 0.975 | 0.960 / 0.990 | 0.977 / 0.973 | 0.973 / 0.974 | 0.972 / 0.972 | 0.978 / 0.975 |
| duration coupling | 0.975 / 0.977 | 0.962 / 0.990 | 0.978 / 0.975 | 0.974 / 0.975 | 0.973 / 0.972 | 0.978 / 0.977 |
| duration marginal | 0.975 / 0.977 | 0.962 / 0.990 | 0.978 / 0.975 | 0.974 / 0.975 | 0.973 / 0.972 | 0.979 / 0.977 |
| rotation | 0.901 / 0.767 | 0.788 / 0.888 | 0.838 / 0.814 | 0.864 / 0.816 | 0.897 / 0.882 | 0.742 / 0.785 |
| jitter $0.25u$ | 0.970 / 0.970 | 0.956 / 0.983 | 0.971 / 0.968 | 0.968 / 0.970 | 0.968 / 0.968 | 0.971 / 0.970 |
| jitter $0.5u$ | 0.959 / 0.957 | 0.945 / 0.971 | 0.956 / 0.958 | 0.955 / 0.958 | 0.955 / 0.960 | 0.957 / 0.958 |
| jitter $1u$ | 0.927 / 0.922 | 0.911 / 0.937 | 0.920 / 0.923 | 0.921 / 0.927 | 0.920 / 0.931 | 0.917 / 0.924 |
| jitter $2u$ | 0.852 / 0.843 | 0.829 / 0.867 | 0.841 / 0.844 | 0.843 / 0.851 | 0.845 / 0.861 | 0.831 / 0.850 |

Table S23: AOI-graph sensitivities on the two halves of a program reading under the default transformations and under support-matched transformations (redrawn until the trial keeps at least its observed dwell share minus 0.02; in brackets, the share of draws for which this succeeded within 200 tries), and the sensitivities of a descriptor made only of the fixation and dwell shares on the support. In the first column: observed AUC of the AOI graph and of the support-only descriptor. Bold: interval excludes zero.

| Problem | Contrast | AOI graph | AOI graph, support-matched | Support-only descriptor |
|---|---|---|---|---|
| `em_halves` (AUC 0.682; 0.773) | rotation | **+0.098 [+0.031, +0.166]** | +0.032 [−0.059, +0.124] (99%) | +0.054 [−0.001, +0.110] |
| `em_halves` (AUC 0.682; 0.773) | jitter $0.25u$ | +0.036 [−0.029, +0.102] | +0.019 [−0.048, +0.086] (100%) | **+0.048 [+0.018, +0.078]** |
| `em_halves` (AUC 0.682; 0.773) | jitter $0.5u$ | +0.023 [−0.025, +0.071] | +0.035 [−0.029, +0.099] (100%) | **+0.112 [+0.065, +0.159]** |
| `em_halves` (AUC 0.682; 0.773) | jitter $1u$ | +0.045 [−0.032, +0.123] | +0.057 [−0.027, +0.141] (89%) | **+0.119 [+0.059, +0.178]** |
| `em_halves` (AUC 0.682; 0.773) | jitter $2u$ | **+0.113 [+0.014, +0.212]** | **+0.121 [+0.035, +0.207]** (31%) | **+0.123 [+0.054, +0.192]** |

Table S24: Duration-marginal transformation of the classic descriptor with the leave-one-participant-out donor pool of the main analysis and with training-fold pools (each fold's trials draw only from that fold's training participants; mean and SD over $R = 6$ replicates; within-fold AUC, class-balanced folds).

| Problem | AUC obs. | AUC, LOO pool | AUC, training-fold pool (SD) | $C$ LOO | $C$ training-fold |
|---|---|---|---|---|---|
| gb_tex_vd1 | 0.969 | 0.935 | 0.935 (0.000) | +0.035 | +0.034 |
| gb_vd1_blg | 0.886 | 0.852 | 0.852 (0.001) | +0.033 | +0.034 |
| gb_vd1_vd2 | 0.640 | 0.516 | 0.523 (0.009) | +0.124 | +0.117 |
| gb_tex_s1s2 | 0.539 | 0.508 | 0.503 (0.020) | +0.031 | +0.036 |
| gb_vd1_halves | 0.564 | 0.556 | 0.553 (0.004) | +0.008 | +0.011 |
| sb_native | 0.689 | 0.672 | 0.672 (0.002) | +0.017 | +0.017 |
| sb_compr | 0.587 | 0.561 | 0.562 (0.005) | +0.026 | +0.024 |
| em_program | 0.539 | 0.507 | 0.532 (0.010) | +0.032 | +0.007 |
| em_expertise | 0.438 | 0.461 | 0.474 (0.038) | −0.023 | −0.037 |
| em_correct | 0.501 | 0.447 | 0.454 (0.023) | +0.054 | +0.047 |
| em_correctrect | 0.523 | 0.474 | 0.411 (0.050) | +0.048 | +0.112 |
| em_correctveh | 0.500 | 0.475 | 0.454 (0.070) | +0.025 | +0.046 |
| em_halves | 0.776 | 0.772 | 0.772 (0.005) | +0.004 | +0.004 |

Table S25: Three permutation distributions for negative sensitivities (within-fold AUC, class-balanced folds; 1000 permutations under the schemes of Table 2 of the article; $R = 6$ transformed replicates). The observed AUC, the transformed AUC and their contrast are each compared with their own permutation distribution; for the contrast the lower-tail probability is given. The first four rows are the order cells of the classic descriptor discussed in Section 5.4 of the article; the last is the unexplained cell of Section 5.11.

| Problem | Descriptor | Transformation | Observed AUC | $p$ | Transformed AUC | $p$ | Contrast | $p$ (lower tail) |
|---|---|---|---|---|---|---|---|---|
| em_program | Classic | order | 0.539 | 0.095 | 0.788 | 0.001 | −0.249 | 0.001 |
| gb_vd1_halves | Classic | order | 0.564 | 0.001 | 0.653 | 0.001 | −0.090 | 0.001 |
| gb_tex_vd1 | Classic | order | 0.969 | 0.001 | 0.991 | 0.001 | −0.022 | 0.142 |
| gb_vd1_blg | Classic | order | 0.886 | 0.001 | 1.000 | 0.001 | −0.114 | 0.001 |
| em_halves | RQA $0.5u$ | jitter $1u$ | 0.613 | 0.001 | 0.692 | 0.001 | −0.080 | 0.027 |

Table S26: Calibration at the largest analysed sample size: 322 participants × 4 trials with the label varying within participants ($R = 6$, $B = 60$; 40 datasets per row; targets from 60 further datasets with $R = 20$, or zero by exchangeability for the order contrast). Coverage of the target by the 95% interval; exclusions of zero where the target is zero by argument; mean width; mean bootstrap SE over the empirical SD of the estimate; coverage of the nominal 80% interval.

| Design | Descriptor | Contrast | Coverage | Excl. 0 at zero target | Width | SE ratio | Coverage at 80% |
|---|---|---|---|---|---|---|---|
| no signal | classic | order | 40/40 | 0/40 | 0.111 | 1.65 | 0.975 |
| no signal | directed | order | 38/40 | 2/40 | 0.139 | 1.39 | 0.950 |
| no signal | directed | jitter 160 px | 39/40 | – | 0.148 | 1.26 | 0.900 |
| no signal | directed | jitter 40 px | 39/40 | – | 0.150 | 1.39 | 0.975 |
| no signal | directed | rotation | 40/40 | – | 0.144 | 1.42 | 0.900 |
| no signal | rqa | order | 40/40 | 0/40 | 0.133 | 1.63 | 0.950 |
| no signal | rqa | jitter 160 px | 40/40 | – | 0.159 | 1.70 | 1.000 |
| no signal | rqa | jitter 40 px | 40/40 | – | 0.150 | 1.62 | 0.925 |
| signal elsewhere | directed | order | 40/40 | 0/40 | 0.024 | 1.38 | 0.900 |
| signal elsewhere | directed | jitter 40 px | 40/40 | – | 0.075 | 1.97 | 1.000 |
| signal elsewhere | directed | rotation | 40/40 | – | 0.045 | 1.44 | 0.975 |
| signal elsewhere | rqa | order | 40/40 | 0/40 | 0.017 | 1.57 | 1.000 |
| signal elsewhere | rqa | jitter 40 px | 40/40 | – | 0.070 | 1.81 | 0.975 |

Table S27: Gradient-boosted trees on the cells of Table 7 of the article at the screening settings and at the long settings (seed 42). Bold: interval excludes zero.

| Problem | Descriptor | Transformation | Trees, $R = 6$, $B = 60$ | Trees, $R = 20$, $B = 300$ |
|---|---|---|---|---|
| gb_tex_vd1 | Dur. trend | duration marginal | **+0.371 [+0.318, +0.424]** | **+0.378 [+0.327, +0.428]** |
| gb_vd1_blg | Classic | order | **−0.118 [−0.148, −0.088]** | **−0.118 [−0.146, −0.090]** |
| gb_vd1_blg | Dur. trend | duration marginal | **+0.226 [+0.161, +0.291]** | **+0.231 [+0.168, +0.293]** |
| gb_vd1_vd2 | Classic | duration marginal | **+0.130 [+0.073, +0.187]** | **+0.127 [+0.074, +0.180]** |
| gb_vd1_vd2 | Dispersion | jitter $2u$ | **+0.102 [+0.038, +0.166]** | **+0.100 [+0.036, +0.164]** |
| gb_vd1_vd2 | Dur. trend | duration coupling | **+0.176 [+0.133, +0.219]** | **+0.172 [+0.128, +0.215]** |
| gb_vd1_vd2 | Dur. trend | duration marginal | **+0.272 [+0.216, +0.329]** | **+0.274 [+0.226, +0.322]** |
| gb_vd1_vd2 | Dur. trend | order | **+0.178 [+0.133, +0.224]** | **+0.175 [+0.132, +0.218]** |
| gb_vd1_vd2 | RQA $0.5u$ | jitter $2u$ | **+0.106 [+0.041, +0.170]** | **+0.103 [+0.037, +0.168]** |
| gb_vd1_vd2 | RQA $1u$ | jitter $2u$ | **+0.125 [+0.054, +0.195]** | **+0.123 [+0.067, +0.178]** |
| gb_vd1_halves | Directed | jitter $2u$ | **+0.192 [+0.140, +0.243]** | **+0.199 [+0.145, +0.252]** |
| gb_vd1_halves | Dispersion | jitter $2u$ | **+0.123 [+0.097, +0.150]** | **+0.124 [+0.096, +0.153]** |
| gb_vd1_halves | Dispersion | rotation | **+0.177 [+0.139, +0.215]** | **+0.167 [+0.133, +0.202]** |
| gb_vd1_halves | RQA $0.5u$ | jitter $2u$ | **+0.157 [+0.101, +0.214]** | **+0.165 [+0.108, +0.222]** |
| gb_vd1_halves | RQA $1u$ | jitter $2u$ | **+0.212 [+0.172, +0.253]** | **+0.219 [+0.168, +0.270]** |
| sb_native | RQA $0.5u$ | jitter $2u$ | −0.002 [−0.084, +0.081] | +0.008 [−0.074, +0.091] |
| sb_native | RQA $1u$ | jitter $2u$ | **+0.100 [+0.009, +0.190]** | **+0.111 [+0.024, +0.199]** |
| sb_native | RQA $1u$ | jitter $1u$ | +0.071 [−0.003, +0.145] | **+0.082 [+0.012, +0.151]** |
| em_program | Dispersion | rotation | **+0.166 [+0.088, +0.244]** | **+0.164 [+0.074, +0.255]** |
| em_halves | Classic | order | **+0.128 [+0.033, +0.222]** | **+0.148 [+0.062, +0.233]** |
| em_halves | RQA $0.5u$ | jitter $1u$ | −0.089 [−0.189, +0.011] | −0.087 [−0.192, +0.018] |

# Methods appendix

## A1. Feature definitions

Let a trial have fixations $(x_i, y_i, d_i)$, $i = 1, \ldots, N$, in sequence order, with $d_i$ the duration.

*Count.* $N$.

*Dispersion.* The population SDs of $x$ and of $y$, and the mean pairwise distance $\binom{N}{2}^{-1} \sum_{i<j} \|p_i - p_j\|$, $p_i = (x_i, y_i)$.

*Classic.* Mean duration $N^{-1} \sum_i d_i$ and mean saccade amplitude $(N-1)^{-1} \sum_{i<N} \|p_{i+1} - p_i\|$.

*Duration trend.* The Pearson correlation of $d_i$ with $i$, and the mean duration; trials with fewer than five fixations are excluded.

*Recurrence* (Anderson et al., 2013). Fixations $i < j$ recur if $\|p_i - p_j\| < r$; $n_r$ is the number of recurrent pairs, and the recurrence plot is the upper triangle of the $N \times N$ recurrence matrix. Recurrence rate: $100 \cdot 2n_r/(N(N-1))$. Determinism: 100 times the share of recurrent pairs that lie on diagonal lines of length at least 2. Laminarity: 100 times the share that lie on vertical lines of length at least 2. Centre of recurrence mass: $100 \cdot \overline{(j-i)}/(N-1)$, the mean over recurrent pairs. A trial without recurrent pairs has all four equal to zero; trials with fewer than five fixations are excluded. Radii: $r = 0.5u$ and $1u$.

*Directed transition graph.* Each fixation is assigned to the tile $(\lfloor x/u \rfloor, \lfloor y/u \rfloor)$; consecutive repeats are collapsed to a sequence $s$ over the visited tiles $V$; $A_{ab}$ counts immediate transitions $a \to b$, and $E = \{(a, b) : a \neq b, A_{ab} > 0\}$. Reciprocity: the share of edges in $E$ whose reverse is in $E$. Revisit tendency: $1 - |V|/|s|$. Transition entropy: $-\sum_{ab} P_{ab} \log_2 P_{ab}$ with $P = A/\sum A$. Degree asymmetry: the mean over tiles of $|\mathrm{out}_a - \mathrm{in}_a|/(\mathrm{out}_a + \mathrm{in}_a)$, with out- and in-strengths the row and column sums of $A$.

*Fixation Adjacency Graph.* Anchors are triangulated (Delaunay). The dwell of a triangle is the summed duration of the fixations inside it; fixations in no triangle are dropped. Vertices are the triangles with positive dwell; two are adjacent if they share an anchor; the weight of an edge is the summed dwell of its two triangles. Attention centrality: the largest, over connected components, of the weight of the component's maximum spanning tree, divided by the total edge weight (set to 0.25 for a graph without edges). Betweenness: the maximum edge betweenness centrality (Brandes, 2001). Compactness: the Estrada index $\sum_k e^{\lambda_k}$ of the adjacency matrix (Estrada, 2000), averaged over components. Squareness: for each component, $\Delta_x \Delta_y / \max(\Delta_x, \Delta_y)^2$ with $\Delta_x, \Delta_y$ the extents of its triangle centroids (zero for a single triangle), averaged over components. The fifth feature is the number of components. Lattice anchors are $2u$ apart and cover the disc swept by the display under rotation; AOI anchors are the token centres of the program; the padded AOI graph adds lattice anchors $2u$ apart around them, leaving out those closer than $2u$ to a token anchor.

## A2. Class sizes and fixation counts

Table S28: Trials and participants per class, and fixations per trial, for every problem (after the exclusions of Section 4 of the article). Class labels are those of the released loaders.

| | | Trials | | Participants | | Fixations per trial | |
|---|---|---|---|---|---|---|---|
| Problem | Positive class | positive | negative | positive | negative | median (IQR) | range |
| `gb_tex_vd1` | `VD1` | 644 | 644 | 322 | 322 | 179 (132–225) | 37–300 |
| `gb_vd1_blg` | `BLG` | 643 | 644 | 322 | 322 | 121 (96–146) | 33–392 |
| `gb_vd1_vd2` | `VD2` | 644 | 644 | 322 | 322 | 141 (121–159) | 32–233 |
| `gb_vd1_halves` | `VD1#2` | 644 | 644 | 322 | 322 | 66 (57–75) | 18–113 |
| `gb_tex_s1s2` | `TEX@S2` | 322 | 322 | 322 | 322 | 225 (206–243) | 97–300 |
| `sb_native` | `nonnative` | 620 | 1426 | 29 | 66 | 122 (92–152) | 7–398 |
| `sb_compr` | `low` | 805 | 558 | 79 | 66 | 121 (93–152) | 8–398 |
| `em_program` | `vehicle` | 112 | 112 | 112 | 112 | 218 (145–291) | 25–868 |
| `em_halves` | `second` | 224 | 224 | 112 | 112 | 109 (72–145) | 12–434 |
| `em_expertise` | `low` | 64 | 51 | 37 | 28 | 181 (129–263) | 25–634 |
| `em_correct` | `incorrect` | 147 | 124 | 120 | 108 | 212 (142–286) | 25–868 |
| `em_correctrect` | `incorrect` | 41 | 92 | 41 | 92 | 193 (142–279) | 35–606 |
| `em_correctveh` | `incorrect` | 106 | 32 | 106 | 32 | 230 (143–293) | 25–868 |

## A3. Software and settings

Python 3.13.16; NumPy 2.5.3; SciPy 1.18.1; pandas 3.0.5; scikit-learn 1.7.2; XGBoost 3.2.0; NetworkX 3.7; package `scanpathdecomp` 0.6.2. Linear SVM: `LinearSVC` (liblinear; squared hinge loss, L2 penalty, $C = 1$, at most 20 000 iterations, random state 0) on features standardized with training-fold means and SDs. Gradient-boosted trees: `XGBClassifier` with 200 trees of depth 3, learning rate 0.05, row subsampling 0.8, histogram tree method, logistic loss, random state 0. Folds: ten participant folds, participants dealt within class in a seeded order (seed 0 for the reported profiles; seeds 0–19 for the partition check).

# References


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